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QUALITY OF PHYSICAL ACTIVITY INFORMATION ON TRADITIONAL WEBSITES VERSUS SOCIAL MEDIA PLATFORMS:

A CROSS-SECTIONAL CONTENT ANALYSIS USING DISCERN AND JAMA BENCHMARKS

FINAL MASTER'S THESIS

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ABSTRACT

QUALITY OF PHYSICAL ACTIVITY INFORMATION ON TRADITIONAL WEBSITES VERSUS SOCIAL MEDIA PLATFORMS: A CROSS-SECTIONAL CONTENT ANALYSIS USING DISCERN AND JAMA BENCHMARKS

FIZINIO AKTYVUMO INFORMACIJOS KOKYBĖ TRADICINĖSE INTERNETO SVETAINĖSE IR SOCIALINIŲ TINKLŲ PLATFORMOSE: SKERSPJŪVIO TURINIO ANALIZĖ TAIKANT DISCERN IR JAMA KRITERIJUS

Keywords: physical activity; health information quality; social media; DISCERN; JAMA benchmarks

Problem. The public increasingly relies on websites and social media for physical activity advice, yet evidence comparing the quality of information across these environments remains limited. Aim. To compare the quality of physical activity information on traditional websites, YouTube, TikTok, and Instagram using DISCERN and JAMA benchmarks. Objectives. To examine platform differences, creator-type differences, and associations between engagement metrics and quality. Hypothesis. Website content and content produced by healthcare or certified fitness professionals would score higher than short-form social media content, whereas engagement metrics would not reliably indicate quality. Methods. A cross-sectional content analysis was conducted on 197 publicly accessible English-language items retrieved with five standardized search terms. Information quality was assessed with the 16-item DISCERN instrument and four JAMA criteria. Inter-rater reliability was evaluated on 40 items. Kruskal-Wallis, Mann-Whitney U, chi-square, and Spearman analyses were used. Main findings. Quality differed significantly by platform: websites ranked highest, followed by YouTube, TikTok, and Instagram. JAMA compliance was low across all platforms. Creator type was associated with DISCERN scores, with healthcare and certified fitness professionals performing better than general users. On YouTube, higher views and likes were associated with lower quality. Conclusions. Traditional websites remain the most reliable source of physical activity information, whereas short-form social media content is more vulnerable to low quality and weak transparency.

SANTRAUKA

FIZINIO AKTYVUMO INFORMACIJOS KOKYBĖ TRADICINĖSE INTERNETO SVETAINĖSE IR SOCIALINIŲ TINKLŲ PLATFORMOSE: SKERSPJŪVIO TURINIO ANALIZĖ TAIKANT DISCERN IR JAMA KRITERIJUS

QUALITY OF PHYSICAL ACTIVITY INFORMATION ON TRADITIONAL WEBSITES VERSUS SOCIAL MEDIA PLATFORMS: A CROSS-SECTIONAL CONTENT ANALYSIS USING DISCERN AND JAMA BENCHMARKS

Raktiniai žodžiai: fizinis aktyvumas; sveikatos informacijos kokybė; socialiniai tinklai; DISCERN; JAMA kriterijai

Problema. Visuomenė vis dažniau fizinio aktyvumo informacijos ieško interneto svetainėse ir socialiniuose tinkluose, tačiau lyginamųjų duomenų apie informacijos kokybę skirtingose skaitmeninėse aplinkose vis dar trūksta. Tikslas. Palyginti fizinio aktyvumo informacijos kokybę tradicinėse interneto svetainėse, YouTube, TikTok ir Instagram, taikant DISCERN instrumentą ir JAMA kriterijus. Uždaviniai. Įvertinti platformų skirtumus, kūrėjų tipų skirtumus ir ryšį tarp įsitraukimo rodiklių bei informacijos kokybės. Hipotezė. Manyta, kad tradicinių interneto svetainių turinys ir sveikatos priežiūros ar sertifikuotų fitneso specialistų sukurtas turinys bus aukštesnės kokybės nei trumpo formato socialinių tinklų turinys, o įsitraukimo rodikliai patikimai neatspindės kokybės. Metodai. Atlikta 197 viešai prieinamų anglų kalba paskelbtų vienetų skerspjūvio turinio analizė, naudojant penkias standartizuotas paieškos frazes. Informacijos kokybė vertinta 16 klausimų DISCERN instrumentu ir keturiais JAMA kriterijais. Tarpvertintojų patikimumas vertintas 40 vienetų imtyje. Taikyti Kruskal-Wallis, Mann-Whitney U, chi kvadrato ir Spearman koreliacijos testai. Pagrindiniai rezultatai. Informacijos kokybė statistiškai reikšmingai skyrėsi tarp platformų: aukščiausius rodiklius pasiekė interneto svetainės, po jų sekė YouTube, TikTok ir Instagram. JAMA kriterijų atitikimas visose platformose buvo žemas. Kūrėjo tipas buvo susijęs su DISCERN įverčiais: geriau pasirodė sveikatos priežiūros ir sertifikuoti fizinio aktyvumo specialistai nei bendrieji naudotojai. YouTube platformoje didesnis peržiūrų ir patiktukų skaičius buvo susijęs su prastesne kokybe. Išvados. Tradicinės interneto svetainės išlieka patikimiausiu fizinio aktyvumo informacijos šaltiniu, o trumpo formato socialinių tinklų turinys dažniau pasižymi prastesne kokybe ir mažesniu skaidrumu.

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LIST OF ABBREVIATIONS

Abbreviation	**Full Term**
APA	American Psychological Association
CI	Confidence Interval
DISCERN	Quality criteria for consumer health information instrument
ICC	Intraclass Correlation Coefficient
IG	Instagram
JAMA	Journal of the American Medical Association
Mdn	Median
MPH	Master of Public Health
PA	Physical Activity
RQ	Research Question
TT	TikTok
WEB	Traditional Website
WHO	World Health Organization
YT	YouTube

INTRODUCTION

Digital media have become a primary route through which the public seeks exercise and physical activity advice. Traditional health websites now coexist with algorithm-driven social media platforms that reward brevity, visual appeal, and engagement. Although this shift expands access to health information, it also creates uncertainty about whether the content most visible to users is accurate, transparent, and useful for health decision-making.

The research problem addressed in this thesis is the uneven and insufficiently compared quality of physical activity information across major online platforms. Previous studies have usually examined single platforms, single health topics, or single assessment tools, which limits direct cross-platform interpretation. As a result, public health practitioners and consumers still lack clear evidence on whether websites, YouTube, TikTok, and Instagram differ in the quality and accountability of physical activity information delivered to ordinary users.

The aim of this thesis is to compare the quality of physical activity information on traditional websites and three major social media platforms using DISCERN and JAMA benchmarks. The objectives are: to assess platform differences in information quality; to examine whether creator type is associated with quality; and to test whether engagement indicators are related to quality. The working hypothesis was that traditional websites and professionally produced content would score higher than short-form social media content, whereas engagement metrics would not reliably signal information quality.

The scientific and practical value of the thesis lies in providing an audited, cross-platform evidence base for digital health communication on physical activity. The findings can support public health guidance, improve health literacy education, and inform platform-level quality interventions. The thesis is structured as follows. Chapter 1 reviews the literature and theoretical background. Chapter 2 presents the methodology. Chapter 3 reports the research findings. Chapter 4 discusses the results, limitations, and implications. The final unnumbered sections present the conclusions, recommendations, references, and annexes.

CHAPTER 1. LITERATURE REVIEW

1.1 Digital Transformation of Health Information Seeking

The emergence of the internet as a primary source of health information represents one of the most consequential shifts in public health communication of the twenty-first century. Where patients once relied almost exclusively on healthcare providers, printed materials, and broadcast media for guidance on medical conditions and health behaviours, the digital era has fundamentally redistributed the locus of health information access (Eysenbach, 2002). This transformation carries profound implications for physical activity (PA) promotion, as individuals increasingly turn to online platforms to learn how, when, and why to exercise. Recent umbrella-review evidence also indicates that digital health interventions can improve lifestyle behaviours, underscoring the practical importance of the information environments through which people encounter exercise advice (Singh et al., 2024).

Recent evidence syntheses confirm that online health information seeking is now a routine part of health management rather than an exceptional behaviour. Jia et al. (2021), in a systematic review of online health information seeking behaviour, concluded that people turn to digital sources because of convenience, speed, anonymity, and the perceived usefulness of web-based information, while patterns of use remain shaped by digital access, trust, and perceived health need. Public-sector communication has also become increasingly digital: Balogun et al. (2023) found that public health agencies relied heavily on social media during pandemics for information dissemination, risk communication, and public engagement, while Afful-Dadzie et al. (2023) highlighted persistent concerns about the quality and credibility of health information circulating through social-media channels.

The digital transformation of health information seeking was catalysed further by the COVID-19 pandemic, which accelerated reliance on digital channels when in-person consultations were restricted (Haghighi & Farhadloo, 2025). A parallel development has been the migration of health information seeking from traditional search engines and institutional

websites to social media platforms. By 2021, 72% of U.S. adults reported ever using at least one social media site (Pew Research Center, 2021), and health-related engagement on these platforms grew in tandem. By 2022, 58.9% of American adults reported watching health-related videos on YouTube, up from 39.7% only two years earlier (Mohamed & Shoufan, 2024). This shift from curated, editorially controlled websites to user-generated content ecosystems has introduced new challenges for information quality assurance.

Eysenbach's (2002) landmark systematic review in JAMA evaluated 79 studies assessing 5,941 health websites and found that 70% of studies concluded that quality was a significant problem on the internet. This foundational finding established the evidence base for ongoing concern about the reliability of digital health information and highlighted the absence of standardised evaluation frameworks at that time. Two decades later, the landscape has grown vastly more complex: the proliferation of social media platforms, the diversification of content formats from text articles to short-form video, and the rise of influencer culture have all compounded the challenge of ensuring that the public receives accurate, evidence-based guidance on health behaviours including physical activity (Suarez-Lledo & Alvarez-Galvez, 2021).

Critically, increased access to digital health information has not been matched by uniformly strong appraisal skills. A recent systematic review and meta-analysis found that eHealth literacy is positively associated with health-related behaviours, indicating that people with stronger digital health appraisal skills are better positioned to act on health information constructively (Kim et al., 2023). Complementary recent syntheses show that eHealth literacy levels vary substantially across populations and are shaped by age, education, and digital access, leaving many users vulnerable when platforms prioritise visibility and engagement over reliability (Jiang et al., 2024; Hua et al., 2025).

1.2 Theoretical Frameworks

Several theoretical frameworks underpin the study of health information quality in digital environments. Together, they provide a multidimensional lens through which the quality of physical activity information across platforms can be understood and assessed.

The Information Quality (IQ) Framework developed by Wang and Strong (1996) constitutes one of the most widely cited conceptualisations of data quality from the consumer perspective. Their empirical study identified 15 quality dimensions organised into four categories: intrinsic quality (accuracy, objectivity, believability, reputation), contextual quality (relevance, value-added, timeliness, completeness, amount of information), representational quality (interpretability, ease of understanding, concise representation, consistent representation), and accessibility quality (accessibility, access security). The framework's emphasis on consumer-defined quality is particularly relevant to health information research, as it shifts the evaluative lens from producer intentions to user needs. When applied to physical activity information online, the IQ Framework highlights that accuracy alone is insufficient; information must also be timely, relevant, interpretable, and accessible to diverse audiences.

Norman and Skinner (2006) extended the concept of health literacy to the digital environment through the development of the eHealth Literacy Scale (eHEALS). They defined eHealth literacy as the ability to seek, find, understand, and appraise health information from electronic sources and apply the knowledge gained to addressing or solving a health problem. The eHEALS is an eight-item self-report measure that assesses consumers' perceived skills and comfort with using digital health resources. Norman and Skinner's model recognises that eHealth literacy is itself a composite construct drawing upon six underlying literacies: traditional literacy, health literacy, information literacy, scientific literacy, media literacy, and computer literacy. This multifaceted conceptualisation is essential for understanding why consumers may struggle to critically evaluate PA information that appears in varying formats across platforms with different interface designs, algorithmic curation strategies, and content norms.

Recent research has refined what lay audiences and patients regard as credible online health information. Daraz et al. (2024) found that people assess credibility using visible signals such as identifiable authorship, transparent sourcing, currency, balance, and institutional trustworthiness. Mohamed et al. (2024) further showed that interventions designed to improve laypeople's ability to identify trustworthy digital health information increasingly focus on strengthening these practical appraisal skills. This contemporary literature reinforces the importance of user-visible quality indicators when evaluating physical activity information across digital platforms.

The Health Belief Model (HBM), originally formulated by Rosenstock (1974) and colleagues at the United States Public Health Service in the 1950s, provides a theoretical basis for understanding why individuals seek health information and how the quality of that information may influence health behaviour. The HBM posits that health behaviour is determined by perceived susceptibility to a health threat, perceived severity of that threat, perceived benefits of taking action, perceived barriers to action, cues to action, and self-efficacy (Rosenstock, 1974). In the context of physical activity, the model suggests that individuals who perceive themselves as at risk of sedentary-behaviour-related health consequences (e.g., cardiovascular disease, obesity) and who believe in the benefits of exercise are more likely to seek PA information online. The quality of the information they encounter may then moderate their perceived self-efficacy and perceived barriers, ultimately influencing whether they adopt and maintain physically active lifestyles. Thus, poor-quality PA information on digital platforms may not merely misinform but may actively undermine the motivational pathways that the HBM describes.

These theoretical perspectives directly inform the study variables used in the present thesis. The Information Quality framework justifies measuring both reliability/completeness (DISCERN) and transparency/accountability (JAMA). eHealth literacy theory explains why users may rely on heuristic cues such as views and likes when appraising online content, providing the rationale for analysing engagement-quality associations. The Health Belief Model clarifies why the quality of PA advice matters for real-world behaviour, especially where uncertainty, risks, and benefits of exercise recommendations are concerned.

1.3 Traditional Websites as Sources of Health and Physical Activity Information

Traditional websites, defined here as content published on the open web and indexed by search engines such as Google, have historically served as the primary digital channel for health information dissemination. These include institutional websites maintained by governmental health agencies (e.g., the World Health Organization, the National Health Service, the Centers for Disease Control and Prevention), professional medical organisations, academic medical centres, patient advocacy groups, and commercial health publishers.

The quality of health information on traditional websites remains highly variable despite the perceived stability of text-based web content. A 2023 systematic review of online nutrition-related information found recurring problems with completeness, sourcing, and accuracy across consumer-facing websites, even when content purported to summarise evidence-based guidance (Denniss et al., 2023). In parallel, recent methodological reviews note substantial variation in how online health information quality is assessed, underscoring the need for transparent and reproducible evaluation criteria when comparing digital sources (Baqraf et al., 2023; Li et al., 2025). These findings suggest that traditional websites may still outperform social media on average, but they should not be assumed to be uniformly trustworthy.

With specific reference to physical activity information, the World Health Organization's (2020) guidelines on physical activity and sedentary behaviour represent the gold standard for evidence-based PA recommendations, prescribing 150-300 minutes of moderate-intensity or 75-150 minutes of vigorous-intensity aerobic activity per week for adults. Institutional websites that accurately reflect these guidelines provide a critical public health resource. However, commercial websites and content aggregators may present PA information that is incomplete, sensationalised, or influenced by commercial interests, potentially leading consumers toward ineffective or unsafe exercise practices.

1.4 Social Media Platforms as Sources of Health Information

The role of social media in health communication has expanded dramatically in recent years. A 2025 systematic review concluded that social media can shape health behaviours through information exposure, social reinforcement, and peer modelling, but also amplify harmful or misleading narratives when engagement incentives dominate (Paul & Headley-Johnson, 2025). Likewise, Afful-Dadzie et al. (2023) found that credibility, completeness, and source transparency remain persistent weaknesses across health communication on social media, despite the reach and accessibility of these platforms.

Each major social media platform possesses distinct characteristics that shape the nature and quality of health content. YouTube, as the world's largest video-sharing platform, supports long-form educational content and has become a major source of exercise tutorials and health information. However, its algorithmic recommendation system prioritises engagement metrics such as views, watch time, and click-through rates, which do not necessarily correlate with accuracy or quality (Li et al., 2024). A 2022 study evaluating YouTube videos recommending exercises during the COVID-19 lockdown found mean DISCERN scores of only 2.29 out of 5 and mean Global Quality Scale (GQS) scores of 2.32, indicating low overall quality (Rodriguez-Rodriguez et al., 2022).

TikTok has emerged as a particularly influential platform for younger demographics, with its short-form video format (typically under 60 seconds) enabling rapid dissemination of health-related content. Li et al. (2024), in a scoping review of 29 studies, found that the quality of most health-related short videos on TikTok was moderate to low, and that DISCERN was the most frequently used evaluation tool across studies, appearing in 24 of the 29 included papers. Critically, the authors noted that most existing quality assessment tools were not designed for short-form video content, raising questions about the validity of applying text-based instruments to multimedia formats. The same review further indicates that medically credentialed creators account for only a minority of health-related TikTok content, with much of the material produced by general users, influencers, and commercial accounts (Li et al., 2024).

Instagram's visual-first architecture, centred on images and short captions with a secondary carousel and Reels format, presents unique challenges for health information quality. Research on Brazilian fitness influencers found that prominent Instagram accounts disseminate low-quality information about exercise and health, with only 2.7% of analysed posts citing any scientific reference (Marocolo et al., 2021). The platform's influencer culture creates commercial and visibility incentives that may conflict with evidence-based health communication. Recent systematic reviews of wellness and physical-activity misinformation on social media indicate that emotionally engaging, simplified, and aspirational messages are often rewarded over well-sourced guidance, especially in lifestyle domains closely tied to identity and appearance (Li & Ye, 2025; Thomas et al., 2025).

The prevalence of health misinformation on social media is well documented. Suarez-Lledo and Alvarez-Galvez (2021), in a systematic review of studies published before March 2019, found that misinformation was most prevalent for content related to smoking products and drugs (reaching 87% in some studies) and vaccines (43%). While physical activity misinformation has received comparatively less attention, the structural incentives for engagement-driven content and the low barriers to publishing health advice on social media platforms suggest that PA information is similarly vulnerable to quality deficits.

1.5 Quality Assessment Tools and Cross-Platform Applicability

Two instruments occupy a central position in the assessment of consumer health information quality: the DISCERN instrument and the JAMA Benchmarks.

DISCERN was developed by Charnock et al. (1999) at the University of Oxford as a standardised tool for judging the quality of written consumer health information about treatment choices. The instrument comprises 16 items: questions 1 through 8 assess the reliability of the publication (Section 1), questions 9 through 15 evaluate the quality of

information about treatment choices (Section 2), and question 16 provides an overall quality rating (Section 3). Each item is scored on a scale from 1 (definite no) to 5 (definite yes), yielding a total score range of 16 to 80. The instrument was validated through a multi-stage process involving expert panels, health information providers, and self-help group members, and has demonstrated acceptable inter-rater reliability (Charnock et al., 1999). DISCERN has subsequently become the most widely used tool for evaluating health information quality, with applications spanning websites, YouTube videos, and social media posts across a broad range of health topics (Li et al., 2024).

The JAMA Benchmarks were proposed by Silberg et al. (1997) as a set of four core standards for health information published on the internet: authorship (identification of authors and their credentials), attribution (clear listing of references and sources), disclosure (transparent declaration of sponsorship, conflicts of interest, and funding), and currency (indication of content creation and update dates). The JAMA Benchmarks are scored as binary criteria (present or absent), yielding a total score of 0 to 4. Their streamlined nature makes them efficient for rapid quality screening, and they complement the more detailed DISCERN assessment by capturing essential transparency indicators.

Recent methodological reviews show that video-based health information studies often combine DISCERN with complementary overall-quality instruments such as the Global Quality Scale, but they also document heterogeneous scoring practices and uneven evidence for measurement properties across tools and platforms (Osman et al., 2022; Li et al., 2025). This heterogeneity is important when interpreting cross-study comparisons of video content quality.

A significant methodological challenge in contemporary health information research is the adaptation of these instruments, all originally developed for text-based content, to multimedia and short-form video formats. Li et al. (2024) noted that most tools used to assess TikTok health videos were not developed for short-form video content and lacked credibility testing in that context. The DISCERN instrument, for example, asks whether the publication refers to areas of uncertainty, a question that may be difficult to operationalise for a 30-second TikTok

video. Similarly, the JAMA authorship criterion requires assessment of whether authors and their credentials are identified, which must be reinterpreted for social media contexts where the creator's bio or verification status serves as a proxy for credentialing. These adaptations, while necessary for cross-platform research, introduce a degree of measurement heterogeneity that must be acknowledged in study design and interpretation. For social media content, researchers have operationalised JAMA criteria such that authorship is satisfied when a creator's bio states professional credentials or the account is a verified professional account, attribution is met when sources are cited in captions or shown on screen, disclosure is met when sponsorships or partnerships are declared, and currency is typically automatically satisfied because social media posts display publication dates.

1.6 Evidence for the Quality Gap and Moderating Factors

A growing body of evidence documents systematic quality differences between health information published on traditional websites and that disseminated through social media platforms. A 2025 meta-analysis published in the Journal of Medical Internet Research, analysing 75 studies encompassing 297,519 posts across 15 platforms, reported a pooled percentage-standardised DISCERN score of 43.58% (95% CI: 37.80–49.35%) for cancer-related information on new media, with approximately 27% of posts containing misinformation (Liu et al., 2025). The analysis further found that studies focusing on video-based media were statistically less likely to yield higher quality conclusions compared with those on text-based media, suggesting a format-dependent quality gradient.

Platform-specific studies reinforce this pattern. Haghighi and Farhadloo (2025), in a cross-sectional analysis of the top 100 health-related websites most frequently shared on Twitter during the early COVID-19 pandemic, found that 81% received low DISCERN scores and 95% met only two of the four JAMA benchmark criteria. No website met all four JAMA criteria, and only five met three. These findings highlight that even the curated content surfacing through social media sharing is of questionable quality. On TikTok, research has consistently demonstrated low overall quality of health content, with the platform's short-

form format constraining the depth and nuance of health information that can be conveyed (Li et al., 2024).

Creator type has emerged as an important moderator of health information quality. Content produced by healthcare professionals and medical organisations generally scores higher on quality assessment tools than content produced by non-medical influencers and general users (Osman et al., 2022; Li et al., 2024). However, the proportion of health content created by qualified professionals varies dramatically across platforms. On TikTok, medically credentialed creators appear to represent only a minority of health information producers, with the remainder largely comprising general users, influencers, and commercial accounts (Li et al., 2024). On Instagram, even creators with relevant academic qualifications rarely cite scientific evidence; Marocolo et al. (2021) found that only 2.7% of posts by major Brazilian fitness influencers included references. This pattern suggests that platform culture and incentive structures may override individual creator qualifications in determining information quality.

The relationship between engagement metrics and information quality represents a particularly concerning dimension of the quality gap. Recent evidence indicates that health misinformation on social media can erode public trust in healthcare and is especially prevalent in wellness and lifestyle domains, where highly engaging content often circulates faster than well-sourced material (Stimpson et al., 2025; Li & Ye, 2025). In the physical-activity domain specifically, Thomas et al. (2025) concluded that misinformation is widespread across social media platforms and highlighted the role of platform affordances in amplifying questionable exercise advice. This creates a structural misalignment between platform incentive systems, which reward engagement, and public health objectives, which require accuracy. For physical activity content specifically, this dynamic may lead to the amplification of extreme exercise regimens, unsupported supplement claims, or unrealistic body-image content at the expense of evidence-based guidelines such as those issued by the World Health Organization (2020).

1.7 Summary and Rationale for the Present Study

The literature reviewed in this chapter establishes several key findings. First, online health information seeking is now a normative behaviour, with the majority of adults regularly consulting digital sources for health guidance, including physical activity information. Second, the migration from traditional websites to social media platforms has introduced new quality challenges, as user-generated content, algorithmic curation, and engagement-driven incentive structures create conditions favourable to the proliferation of low-quality and misleading health information. Third, established quality assessment instruments, particularly DISCERN and the JAMA Benchmarks, provide reliable and validated tools for evaluating health information, though their application to multimedia and short-form video content requires careful operationalisation. Fourth, the existing evidence consistently indicates that health information quality on social media is generally lower than on traditional websites, with substantial variability by platform type, creator qualifications, and content format.

Despite the breadth of this literature, a significant gap remains. To date, no single study has conducted a head-to-head comparison of physical activity information quality across all four major platform types, namely traditional websites, YouTube, TikTok, and Instagram, using both DISCERN and JAMA Benchmarks within a unified methodological framework. Existing studies have typically examined one or two platforms, focused on a specific health condition rather than general physical activity content, or used only one quality assessment instrument. The absence of a comprehensive, multi-platform comparison limits the ability to draw actionable conclusions about where the greatest quality deficits lie and what factors, such as creator type and engagement metrics, may moderate quality across the digital information ecosystem.

The present study addresses this gap by conducting a cross-sectional content analysis of 197 items across traditional websites, YouTube, TikTok, and Instagram, applying both the DISCERN instrument and JAMA Benchmarks to assess physical activity information quality. The final audited sample comprised 50 websites, 50 YouTube videos, 47 TikTok videos, and 50 Instagram posts. The study is guided by three research questions:

1. How does physical activity information quality differ between traditional websites and social media platforms as measured by DISCERN and JAMA benchmarks?

2. Does creator type (healthcare professional versus fitness influencer versus general user) moderate quality scores?

3. What is the relationship between engagement metrics (views, likes, shares) and information quality?

By addressing these questions within a single, standardised methodological framework, this study aims to provide the most comprehensive cross-platform assessment of physical activity information quality to date, generating evidence that can inform platform-level interventions, health literacy initiatives, and clinical guidance on directing patients toward reliable digital PA resources.

CHAPTER 2. RESEARCH METHODOLOGY

This chapter presents the methodological framework employed to investigate the quality of physical activity (PA) information across traditional websites and social media platforms. The research methodology encompasses the research object, strategy and logic, nature of the research, description of research subjects, data collection and assessment methods, organisational procedures, and statistical analysis techniques. Each component is detailed to ensure transparency, replicability, and methodological rigour in alignment with established standards for content analysis in health communication research.

2.1 Research Object

The research object of the present study is the quality of physical activity information available on traditional websites and social media platforms. Specifically, this investigation examines health-related content pertaining to physical activity that is publicly accessible through Google search results (representing traditional web-based information), YouTube, TikTok, and Instagram (representing contemporary social media platforms). Physical activity information is defined as any content that provides guidance, recommendations, instructions, or educational material regarding exercise, fitness, or physical activity behaviours intended for a general adult audience.

The assessment of information quality is operationalised through two established benchmarks: the DISCERN instrument (Charnock et al., 1999), which evaluates the reliability and completeness of health information, and the JAMA benchmarks (Silberg et al., 1997), which assess accountability standards including authorship, attribution, disclosure, and currency. Together, these instruments provide a comprehensive evaluation framework that captures both the substantive quality and the transparency of health information across diverse digital platforms.

2.2 Research Strategy and Logic

The present study employs a cross-sectional content analysis design, which enables the systematic examination and comparison of physical activity information quality across four digital platforms at a single point in time. Content analysis is a well-established research method in health communication that facilitates the objective, systematic, and quantitative description of manifest content (Krippendorff, 2018). The cross-sectional approach was

selected for its capacity to capture a snapshot of the current information landscape, allowing for direct comparisons across platforms under identical assessment conditions.

The research follows a deductive approach, applying established quality assessment frameworks—the DISCERN instrument and JAMA benchmarks—to evaluate content across a novel four-platform comparison. While these instruments were originally developed for traditional health information materials, they are widely used in studies of YouTube and TikTok health content despite recognised methodological concerns when applied to audiovisual and short-form media (Azer, 2020; Kong et al., 2021; Li et al., 2024). The present study therefore treats their use as a transparent, controlled adaptation rather than as a platform-specific validation.

The study design was structured using the SPIDER framework (Cooke et al., 2012), adapted for content analysis as follows:

- **Sample:** Targeted sample of 200 content items across four platforms; final audited analytical sample of 197 items after three unverifiable TikTok records were excluded.
- **Phenomenon of Interest:** Quality of physical activity information.
- **Design:** Cross-sectional content analysis.
- **Evaluation:** DISCERN instrument and JAMA benchmarks.
- **Research type:** Quantitative.

The rationale for employing a cross-sectional content analysis rather than a systematic review lies in the current state of the literature. Recent systematic reviews and meta-analyses have already synthesised major strands of evidence on health information quality across websites, social media, and short-form video content (Denniss et al., 2023; Li et al., 2024; Liu et al., 2025). However, no study to date has conducted a direct, head-to-head comparison of physical activity information quality across traditional websites, YouTube, TikTok, and Instagram using both the DISCERN and JAMA instruments simultaneously. The present study addresses this gap by providing a four-platform comparative analysis using dual quality assessment frameworks applied specifically to physical activity content.

2.3 Nature of Research

The present study is quantitative in nature, employing numerical scoring instruments to assess information quality and statistical methods to analyse the resulting data. The research is both descriptive and comparative: descriptive in that it characterises the quality of

physical activity information across platforms using standardised metrics, and comparative in that it systematically examines differences in quality scores between platform types, creator categories, and content formats.

Furthermore, this study represents applied research with practical implications for multiple stakeholders. The findings are intended to inform health literacy initiatives, guide content regulation policies on social media platforms, assist healthcare professionals in directing patients toward reliable information sources, and contribute to the broader understanding of digital health information ecosystems. The applied orientation aligns with recent evidence showing that social-media-based health information can shape health behaviours while misinformation in these environments can undermine public trust and decision-making (Paul & Headley-Johnson, 2025; Stimpson et al., 2025).

2.4 Study Sample and Sampling Frame

The sampling design targeted 200 content items from four digital platforms: traditional websites accessed through Google search (target n = 50), YouTube (target n = 50), TikTok (target n = 50), and Instagram (target n = 50). This target sample size was designed to provide balanced cross-platform coverage and is comparable to sample sizes commonly used in recent health-information quality studies and reviews of video-based and short-form social media content (Osman et al., 2022; Li et al., 2024). Following the post-collection audit, three TikTok records could not be verified against the raw captures and were excluded without replacement, yielding a final analytical sample of 197 items (50 websites, 50 YouTube videos, 47 TikTok videos, and 50 Instagram posts).

Verification required a raw screenshot or captured record showing the content identifier or URL together with the metadata fields used in analysis (for example, platform, creator, date, content format, or engagement counts). Records that could not be matched to this audit evidence were removed from the final audited dataset.

Content was retrieved using five standardised search terms designed to simulate typical user queries related to physical activity:

- “How to start exercising”
- “Best exercises to lose weight”
- “Strength training for beginners”
- “Physical activity guidelines”

• “Home workout routine”

These search terms were selected to represent a range of common physical activity information-seeking behaviours, from general exercise initiation to specific activity modalities. For each search term, the first 10 results were targeted from each platform, yielding a planned collection frame of 10 results × 5 search terms × 4 platforms = 200 items. The sampling strategy employed convenience sampling of top-ranked search results, which approximates the actual information-seeking behaviour of typical users who predominantly access the first page or highest-ranked results (Pan et al., 2007). After the audit exclusions described above, the final analytical sample comprised 197 items.

Inclusion criteria. Content items were included if they met all of the following criteria: (a) written or presented in English, (b) publicly accessible without account registration or payment, (c) containing substantive physical activity information (i.e., providing guidance, instructions, or educational content about exercise or physical activity), and (d) posted or last updated within the three years preceding data collection (i.e., from February 2023 onward).

Exclusion criteria. Content items were excluded if they met any of the following criteria: (a) duplicate content appearing across multiple platforms, (b) content behind a paywall or requiring subscription access, (c) content not primarily focused on physical activity (e.g., product-only advertisements, unrelated entertainment), or (d) content in languages other than English. During initial collection, excluded search results were replaced by the next sequential organic result from the same search to preserve the planned sampling frame. During the later audit, three TikTok records lacked verifiable raw-source support and were therefore removed without replacement.

Accordingly, Table 2.1 should be interpreted as the planned sampling matrix rather than the final audited analytical distribution. All inferential analyses reported in Chapters 3–4, the concluding sections, and the appendices are based on the final audited dataset of 197 items.

Table 2.1 presents the sampling matrix illustrating the distribution of content items across search terms and platforms.

Table 2.1

Planned Sampling Matrix: Target Distribution of Content Items by Search Term and Platform

Search Term	Google (n)	YouTube (n)	TikTok (n)	Instagram (n)	Total (N)
How to start exercising	10	10	10	10	40
Best exercises to lose weight	10	10	10	10	40
Strength training for beginners	10	10	10	10	40
Physical activity guidelines	10	10	10	10	40
Home workout routine	10	10	10	10	40
Total	**50**	**50**	**50**	**50**	**200**

Note. Each cell represents the targeted number of content items per search term per platform. Planned collection frame N = 200; final audited analytical sample N = 197 after three TikTok exclusions.

2.5 Research Methods

2.5.1 Data Collection Procedures

Data collection was conducted over a two-day window (1–2 March 2026) using a systematic protocol designed to reduce avoidable sources of algorithmic bias and improve reproducibility. All searches were performed in logged-out, incognito (private) browsing mode in Google Chrome to reduce the influence of browsing history, cookies, and account-based recommendations on result rankings (Tripodi, 2018). Nonetheless, search results may still vary because of location/IP-based and other forms of platform personalisation. The browser's language was set to English (United States) and geolocation was configured to Kaunas, Lithuania, the researcher's institutional location, to standardise the search environment across all platforms. No VPN was used.

For traditional websites, searches were conducted on Google (google.com) using the default 'All' results tab; sponsored results and advertisements were excluded. For YouTube, searches were conducted on youtube.com using the default 'Relevance' sort. For both platforms, an automated data collection approach was employed using Playwright, an open-source browser automation framework. The automated scripts navigated to each platform, entered the standardised search terms, and systematically recorded the top 10 organic results for each query. This approach ensured consistency in result capture and reduced human error in the recording of search rankings, URLs, and metadata. For TikTok and Instagram, manual data collection was conducted due to these platforms' implementation of anti-bot measures that prevent reliable browser automation. TikTok searches used the in-app 'Top' tab; Instagram searches used the 'Tags' and 'Top' tabs. The primary researcher manually searched each term within the platform's native search function, recording the first 10 relevant results in order of appearance. For TikTok and Instagram, this order of appearance was treated as an operational proxy for ranking position rather than a guaranteed ordinal rank

list. Reposts, advertisements, pinned/promoted content, and duplicate items from the same creator for the same search term were excluded and replaced by the next sequential organic result during the initial collection stage.

All search results were recorded in their order of appearance within the search ranking. For each content item, the following information was documented: platform, search term used, ranking position, URL or content identifier, creator name, creator type classification, content format, date of publication or last update, and engagement metrics (views, likes, shares, and comments where available). Screenshots were captured for each content item to establish an audit trail, supporting the verification and reliability of the assessment process.

2.5.2 Quality Assessment Instruments

DISCERN Instrument. The DISCERN instrument (Charnock et al., 1999) is a validated and widely used tool for assessing the quality of written consumer health information. The instrument comprises 16 items, each scored on a Likert-type scale ranging from 1 (definite no) to 5 (definite yes). The instrument is divided into three sections: Section 1 (Questions 1–8) evaluates the reliability of the publication, addressing whether the content has clear aims, achieves those aims, is relevant, identifies its sources, is balanced, and provides sufficient detail for decision-making. Section 2 (Questions 9–15) assesses the quality of information about treatment choices, including descriptions of how treatments work, their benefits, risks, effects on quality of life, and what would happen without treatment. Section 3 comprises a single item (Question 16) providing an overall quality rating. Total scores range from 16 to 80.

For the purposes of the present study, DISCERN quality scores were categorised using the following established thresholds: Very Poor (16–26), Poor (27–38), Fair (39–50), Good (51–62), and Excellent (63–80). These conventional categories facilitate meaningful interpretation and cross-study comparison of quality ratings (Sun et al., 2023). In adapting the instrument for social media content, questions pertaining to treatment information were interpreted in the context of exercise and physical activity recommendations, consistent with recent studies examining health-related video and short-form social-media content (Kong et al., 2021; Li et al., 2024).

Scoring was restricted to information visible to a typical user at the point of access. For websites, raters assessed the full article text and any immediately visible source, date, and

authorship information. For YouTube and TikTok, raters were permitted to use the video content itself, spoken narration, on-screen text, title, caption/description, pinned comments where present, and creator profile bio. For Instagram, raters used all visible carousel text or image text, the caption, pinned comments where present, and profile bio. No score was awarded on the basis of inferred but unstated evidence or credentials; if a risk, source, or qualification was implied but not explicitly visible, the conservative lower score was retained.

Table 2.2 presents the complete DISCERN instrument with all 16 assessment questions.

Table 2.2

DISCERN Instrument Assessment Questions

Item	Section	Question
Q1	1: Reliability	Are the aims clear?
Q2	1: Reliability	Does it achieve its aims?
Q3	1: Reliability	Is it relevant?
Q4	1: Reliability	Is it clear what sources of information were used to compile the publication?
Q5	1: Reliability	Is it clear when the information used or reported in the publication was produced?
Q6	1: Reliability	Is it balanced and unbiased?
Q7	1: Reliability	Does it provide details of additional sources of support and information?
Q8	1: Reliability	Does it refer to areas of uncertainty?
Q9	2: Treatment Quality	Does it describe how each treatment works?
Q10	2: Treatment Quality	Does it describe the benefits of each treatment?
Q11	2: Treatment Quality	Does it describe the risks of each treatment?
Q12	2: Treatment Quality	Does it describe what would happen if no treatment is used?
Q13	2: Treatment Quality	Does it describe the effects of treatment choices on overall quality of life?
Q14	2: Treatment Quality	Is it clear that there may be more than one possible treatment choice?
Q15	2: Treatment Quality	Does it provide support for shared decision-making?
Q16	3: Overall Rating	Based on the answers to all of the above questions, rate the overall quality of the publication.

Note. Adapted from "DISCERN: An instrument for judging the quality of written consumer health information on treatment choices," by D. Charnock, S. Shepperd, G. Needham, and R. Gann, 1999, Journal of Epidemiology & Community Health, 53(2), 105–111. Each item is scored from 1 (definite no) to 5 (definite yes). For social media content, treatment-related questions (Q9–Q15) were interpreted as referring to exercise and physical activity recommendations.

JAMA Benchmarks. The JAMA benchmarks (Silberg et al., 1997) provide a concise accountability framework for evaluating health information. The instrument assesses four criteria, each scored dichotomously as present (Y) or absent (N): (a) Authorship—whether

authors and their credentials or affiliations are clearly identified; (b) Attribution—whether references and sources for content are cited; (c) Disclosure—whether sponsorship, advertising, conflicts of interest, or financial relationships are declared; and (d) Currency—whether the date of content creation or most recent update is provided. The total JAMA score ranges from 0 to 4, representing the number of criteria met.

To apply the JAMA benchmarks to social media content, which differs substantially in format from traditional web-based health information, the criteria were operationalised as described in Table 2.3. This operationalisation draws on approaches reported in recent research and reviews assessing YouTube and TikTok health content (Osman et al., 2022; Li et al., 2024).

Table 2.3

JAMA Benchmarks Criteria and Social Media Operationalisation

Criterion	Original Definition	Social Media Operationalisation
Authorship	Authors and credentials/affiliations identified	Creator bio states professional credentials (e.g., MD, PT, CSCS) or account is a verified professional/organisational account
Attribution	References and sources for content are cited	Content cites scientific studies, clinical guidelines, or other sources in video, caption, description, or on-screen text references
Disclosure	Sponsorship, advertising, or conflicts of interest declared	Content discloses sponsorships, partnerships, or paid promotions (e.g., #ad, #sponsored, platform-provided paid partnership label)
Currency	Date of content creation or update provided	Post or publication date is visible (inherent feature of most social media platforms; for websites, date must be explicitly displayed)

Note. Adapted from "Assessing, controlling, and assuring the quality of medical information on the internet," by W. M. Silberg, G. D. Lundberg, and R. A. Musacchio, 1997, JAMA, 277(15), 1244–1245. Each criterion is scored as present (Y) or absent (N). CSCS = Certified Strength and Conditioning Specialist; MD = Medical Doctor; PT = Physical Therapist.

2.5.3 Additional Variables

In addition to the primary quality assessment instruments, several supplementary variables were recorded for each content item to facilitate secondary analyses examining the moderating role of creator characteristics and content features on information quality.

Creator type classification. Each content creator was classified into one of five categories based on the information available in their profile, bio, and content: (a) Healthcare professional (e.g., physicians, physiotherapists, registered nurses), (b) Certified fitness professional (e.g., certified personal trainers, strength and conditioning specialists), (c)

Fitness influencer (individuals with substantial followings who create fitness content without verified professional credentials), (d) General user (individuals without identifiable fitness or health credentials), and (e) Organisation (institutional accounts such as hospitals, health agencies, or fitness organisations). This classification scheme was informed by recent reviews of creator typologies in YouTube, TikTok, and broader social-media health content (Osman et al., 2022; Li et al., 2024; Thomas et al., 2025).

Engagement metrics. Where available, the following engagement metrics were recorded at the time of data collection: total views, likes (or equivalent reactions), shares, and comments. These metrics serve as indicators of content reach and audience interaction, enabling the examination of the relationship between content popularity and information quality (Osman et al., 2022).

Content format. Each content item was classified by format: (a) Text article (traditional websites), (b) Long-form video (>5 minutes; primarily YouTube), (c) Short-form video (<60 seconds; primarily TikTok and Instagram Reels), (d) Image with caption (primarily Instagram posts), and (e) Carousel (multi-image Instagram posts). This classification enables the examination of whether content format is associated with information quality.

2.6 Research Organisation

The study was conducted over a defined period from February to March 2026, with data collection completed within a concentrated two-day window to minimise temporal variation in search results and engagement metrics. The research process was organised into six sequential phases, as illustrated in Figure 2.1.

Phase 1: Automated data collection (Day 1). Playwright browser automation scripts were executed to collect search results from Google and YouTube. The scripts performed searches for all five search terms, captured the top 10 results per term, and recorded URLs, metadata, and screenshots.

Phase 2: Manual data collection (Days 1–2). The primary researcher manually collected search results from TikTok and Instagram using the platform-native search functions. Results were recorded in the same standardised format as the automated collection.

Pilot scoring. Prior to the full assessment, the primary researcher conducted a pilot scoring exercise on 10 content items (2–3 per platform, not included in the final sample) to

test the operationalisation of the DISCERN and JAMA instruments for social media content. This pilot identified that DISCERN questions Q4 (sources of information) and Q11 (risks of treatment) were the most challenging to adapt to short-form video, as these items often required careful distinction between explicitly visible evidence and implied claims. Scoring rules were refined accordingly: for Q4, on-screen citation of sources or verbal references to guidelines were accepted; for Q11, any explicit mention of injury risk, contraindications, or the advice to consult a professional was scored positively. These pilot-derived rules were documented in the scoring guide and applied consistently across the final audited dataset.

Phase 3: Primary quality assessment. The primary researcher assessed all 197 audited content items using both the DISCERN instrument and the JAMA benchmarks. For text-based website content, each article was read in full. For video content (YouTube and TikTok), scoring was based on a combination of the audiovisual content, visible captions, descriptions, on-screen text, and profile information; full verbatim transcription was not performed. For Instagram carousels and image posts, scoring was based on all visible text, captions, and associated profile information. This approach reflects the information a typical user would encounter when engaging with each item. Scores were recorded in a structured data collection spreadsheet.

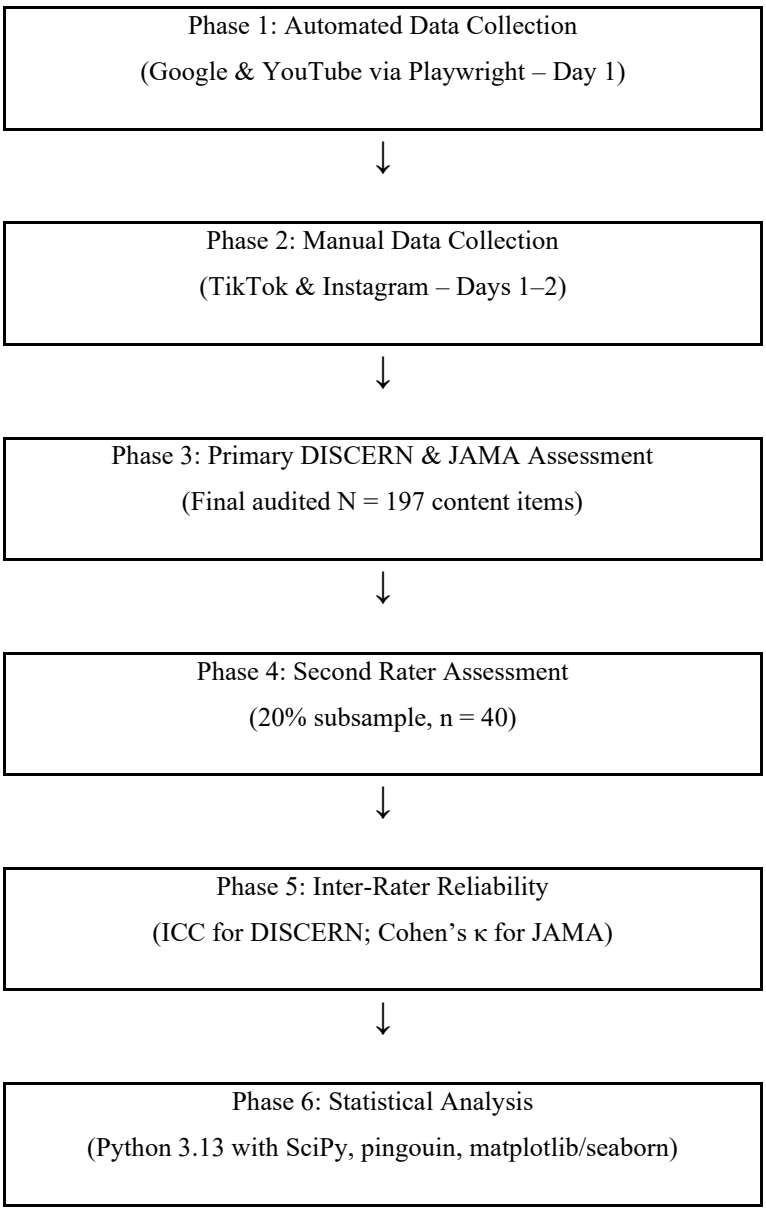
Phase 4: Second rater assessment. A trained second rater—a fellow Master of Public Health student with prior coursework in epidemiology and health communication—independently scored a 20% subsample of content items (n = 40; 10 randomly selected items per platform) using both instruments. Prior to independent scoring, the second rater received a two-hour training session covering the DISCERN instrument, JAMA benchmarks, and their social media operationalisations, followed by supervised practice scoring of five items (not included in the final sample) with discussion of discrepancies. The second rater was then provided with the same scoring guidelines, quick-reference card, and worked examples as the primary researcher to ensure standardised application of the assessment criteria.

Phase 5: Inter-rater reliability assessment. Agreement between the primary researcher and the second rater was evaluated using the Intraclass Correlation Coefficient (ICC) for DISCERN scores and Cohen's kappa (κ) for JAMA criteria. These metrics are reported in the Results chapter.

Phase 6: Statistical analysis. Following the completion of quality assessments, all data were compiled and analysed using Python 3.13 with scientific computing libraries, as detailed in Section 2.7.

Figure 2.1

Flow Diagram of the Research Process



Note. The research process proceeded sequentially through six phases over the study period (February–March 2026).

2.6.1 Ethical Considerations

The present study did not involve direct interaction with human participants, as all data were derived from publicly available online content. Consequently, formal ethics

committee approval was not required, consistent with standard practice for content analyses of publicly accessible materials (Snelson, 2016). Nonetheless, the study was designed in line with internet research ethics principles by limiting data collection to public-facing content, avoiding private groups or restricted materials, not bypassing access controls, and minimising unnecessary identification of individuals in the thesis text. No personal data were collected beyond publicly displayed profile information relevant to creator-type classification, and screenshots were retained solely as an audit trail for verification purposes.

2.7 Methods of Statistical Analysis

All statistical analyses were conducted using Python 3.13 with the following scientific computing libraries: pandas (McKinney, 2010) for data manipulation, SciPy (Virtanen et al., 2020) for non-parametric inferential tests including Kruskal–Wallis, Mann–Whitney U, chi-square, and Spearman correlations, pingouin (Vallat, 2018) for intraclass correlation coefficient computation, and matplotlib/seaborn for publication-quality graphical representations. The significance level was set at $\alpha = 0.05$ for all inferential tests. The complete analysis code is reproduced in Appendix D.

2.7.1 Descriptive Statistics

Descriptive statistics were employed to characterise the distribution of quality scores across platforms and content categories. For DISCERN total scores, the median, interquartile range (IQR), and range were reported for each platform, as ordinal Likert-type data are best represented by median-based measures of central tendency (Sullivan & Artino, 2013). For JAMA benchmarks, frequencies and percentages of compliance for each criterion were calculated per platform, along with overall JAMA score distributions.

2.7.2 Inter-Rater Reliability

Inter-rater reliability was assessed for the 20% subsample ($n = 40$) independently scored by both the primary researcher and the second rater. For DISCERN total scores, the Intraclass Correlation Coefficient (ICC) was calculated using a two-way mixed-effects model, single measures, with absolute agreement (ICC[3,1]; Koo & Li, 2016). ICC values were interpreted according to established guidelines: values less than 0.50 indicate poor reliability, 0.50–0.75 indicate moderate reliability, 0.75–0.90 indicate good reliability, and values greater than 0.90 indicate excellent reliability (Koo & Li, 2016).

For the JAMA benchmark criteria, which are scored dichotomously (present/absent), Cohen's kappa (κ) was calculated for each criterion. Kappa values were interpreted using the guidelines proposed by Landis and Koch (1977): values below 0.20 indicate slight agreement, 0.21–0.40 indicate fair agreement, 0.41–0.60 indicate moderate agreement, 0.61–0.80 indicate substantial agreement, and values above 0.80 indicate almost perfect agreement.

2.7.3 Inferential Statistics

Non-parametric statistical tests were selected as the primary inferential methods due to the ordinal nature of DISCERN scores and the anticipated non-normal distribution of quality data, consistent with methodological recommendations for analysing Likert-type rating scales (Sullivan & Artino, 2013).

Platform comparisons of DISCERN scores. The Kruskal-Wallis H test was employed to examine whether DISCERN total scores differed significantly across the four platforms (websites, YouTube, TikTok, and Instagram). When a statistically significant omnibus result was obtained, pairwise Mann–Whitney U tests with Bonferroni correction were applied to identify which specific platform pairs differed significantly. The omnibus effect size was computed using the formula $(H - k + 1) / (N - k)$ and is reported as an eta-squared approximation for Kruskal-Wallis comparisons. This value should be interpreted cautiously as an index of distributional separation rather than a literal proportion of explained variance; values of approximately 0.01, 0.06, and 0.14 correspond to small, medium, and large effects, respectively (Cohen, 1988).

Platform comparisons of JAMA compliance. The chi-square test of independence was used to assess whether the proportion of content meeting each JAMA criterion differed across platforms. Where expected cell frequencies fell below 5, Fisher's exact test was employed as an alternative. The effect size was reported as Cramér's V, interpreted according to Cohen's (1988) guidelines: values of 0.10, 0.30, and 0.50 represent small, medium, and large effects for tables with three degrees of freedom.

Creator type and quality scores. To investigate whether creator type moderates information quality, DISCERN scores were compared between professional creators (healthcare professionals and certified fitness professionals combined) and non-professional creators (fitness influencers, general users, and organisations combined) using the Mann-Whitney U test. Organisations were classified as non-professional because they encompass a heterogeneous mix of entities—from government health departments to commercial fitness

brands—making uniform professional classification inappropriate. The effect size was reported as rank-biserial correlation (r_rb), where values of 0.10, 0.30, and 0.50 correspond to small, medium, and large effects (Fritz et al., 2012).

Engagement metrics and information quality. Spearman's rank correlation coefficient (ρ) was calculated to examine the relationship between engagement metrics (views, likes, shares) and DISCERN total scores. This non-parametric correlation was selected due to the highly skewed distribution typically observed in engagement data and the ordinal nature of DISCERN scores. Correlation strength was interpreted as: $\rho < 0.10$ negligible, 0.10–0.39 weak, 0.40–0.69 moderate, and ≥ 0.70 strong (Schober et al., 2018).

Table 2.4 provides a summary of the statistical analysis plan, linking each research question to the corresponding variables, statistical tests, and rationale for test selection.

Table 2.4

Summary of Statistical Analysis Plan by Research Question

Research Question	Variables	Statistical Test	Rationale
RQ1: How does PA information quality differ between traditional websites and social media?	IV: Platform (4 levels) DV: DISCERN total score; JAMA criteria compliance	Kruskal-Wallis H test (DISCERN) Pairwise Mann-Whitney U with Bonferroni χ^2 / Fisher's exact test (JAMA) Effect sizes: eta-squared approximation, Cramér's V	Non-parametric; ordinal DISCERN data likely non-normally distributed. Chi-square appropriate for categorical JAMA data.
RQ2: Does creator type moderate quality scores?	IV: Creator type (professional vs. non-professional) DV: DISCERN total score	Mann-Whitney U test Effect size: rank-biserial correlation	Non-parametric comparison of two independent groups with ordinal outcome variable.
RQ3: What is the relationship between engagement metrics and quality?	Predictor: Views, likes, shares Outcome: DISCERN total score	Spearman's rank correlation (ρ)	Non-parametric correlation appropriate for skewed engagement data and ordinal quality scores.
Inter-rater reliability	Rater 1 vs. Rater 2 scores (20% subsample, n = 40)	ICC (two-way mixed, single measures, absolute agreement) for DISCERN Cohen's κ for each JAMA criterion	ICC suitable for continuous/ordinal data; κ appropriate for dichotomous categorical data.

Note. IV = independent variable; DV = dependent variable; PA = physical activity; ICC = Intraclass Correlation Coefficient; χ^2 = chi-square; κ = kappa; ρ = Spearman's rho. The Kruskal-Wallis omnibus effect size was calculated as $(H - k + 1) / (N - k)$. Significance level: $\alpha = 0.05$ for all tests.

In summary, the methodological approach described in this chapter was designed to provide a rigorous, transparent, and replicable framework for evaluating the quality of physical activity information across four major digital platforms. The combination of the

DISCERN instrument and JAMA benchmarks offers a comprehensive assessment of both substantive information quality and accountability standards. The use of non-parametric statistical methods is appropriate given the ordinal nature of the primary outcome measures, and the inclusion of inter-rater reliability assessment ensures the credibility of the quality ratings. The results of the analyses described herein are presented in Chapter 3.

CHAPTER 3. RESEARCH FINDINGS

3.1 Sample Characteristics

A total of 197 items were included in the final analysis across four platforms: traditional websites (n = 50), YouTube (n = 50), TikTok (n = 47), and Instagram (n = 50). Each platform was initially targeted for 10 results per search term; however, three TikTok items were excluded following a data audit, yielding a TikTok subsample of 47. All 197 items met the predefined inclusion criteria of being publicly accessible, English-language, substantive physical activity content published or updated within the past three years.

Creator types were classified into five categories. As shown in Table 1, organisations constituted the largest group (n = 84, 42.6%), followed by fitness influencers (n = 38, 19.3%), certified fitness professionals (n = 32, 16.2%), general users (n = 24, 12.2%), and healthcare professionals (n = 19, 9.6%). Organisations were particularly prevalent among traditional websites (n = 36, 72.0%) and Instagram (n = 29, 58.0%), reflecting the dominance of institutional content on these platforms. In contrast, TikTok was characterised by greater diversity in creator types, with the highest proportions of fitness influencers (n = 19, 40.4%) and general users (n = 9, 19.1%).

Table 1. Distribution of Creator Types by Platform

Creator Type	Website	YouTube	TikTok	Instagram	Total
Healthcare Professional	5 (10%)	3 (6%)	8 (17%)	3 (6%)	19 (9.6%)
Certified Fitness Prof.	5 (10%)	13 (26%)	6 (13%)	8 (16%)	32 (16.2%)
Fitness Influencer	0 (0%)	16 (32%)	19 (40%)	3 (6%)	38 (19.3%)
General User	4 (8%)	4 (8%)	9 (19%)	7 (14%)	24 (12.2%)
Organisation	36 (72%)	14 (28%)	5 (11%)	29 (58%)	84 (42.6%)
Total	50	50	47	50	197

Content formats were entirely platform-specific (Table 2). Following a data audit, each platform was found to be exclusively associated with a single content format. Traditional websites comprised text articles only (n = 50, 100%). YouTube consisted entirely of long-form videos exceeding five minutes (n = 50, 100%). TikTok content was exclusively short videos under 60 seconds (n = 47, 100%). Instagram comprised only carousel posts (n = 50, 100%). This clean one-to-one correspondence between platform and format means that content format and platform are perfectly confounded in this dataset; format differences in

quality scores therefore reflect platform differences and cannot be attributed to format independently of platform.

Table 2. Distribution of Content Formats by Platform

Platform	Content Format	n (%)
Website	Text Article	50 (100%)
YouTube	Long Video (>5 min)	50 (100%)
TikTok	Short Video (<60 s)	47 (100%)
Instagram	Carousel	50 (100%)

3.2 Inter-Rater Reliability

A second rater independently scored a random subset of 40 items (20% of the total sample; 10 per platform) to establish inter-rater reliability. For DISCERN total scores, the intraclass correlation coefficient (ICC) was calculated using a two-way mixed-effects model with absolute agreement (ICC[3,1]). The resulting ICC was 0.932 (95% CI [0.870, 0.960]), indicating excellent agreement between raters (Koo & Li, 2016). This value substantially exceeds the pre-specified threshold of ICC >= 0.75.

For the four binary JAMA benchmark criteria, inter-rater agreement was assessed using Cohen's kappa. Authorship (kappa = 0.719) and Attribution (kappa = 0.686) demonstrated substantial agreement (Landis & Koch, 1977). However, Disclosure (kappa = 0.305) and Currency (kappa = 0.215) yielded only fair agreement. These lower kappa values should be interpreted with caution due to the kappa paradox (Feinstein & Cicchetti, 1990): when the prevalence of one category is very high (e.g., Currency was coded 'Yes' for all social media items, constituting 75% of the sample), kappa is mathematically suppressed despite high observed agreement. The overall percentage agreement for Currency was 92.5% and for Disclosure was 87.5%, suggesting that the low kappa values reflect distributional artefacts rather than genuine disagreement between raters.

3.3 DISCERN Scores by Platform

Descriptive statistics for DISCERN total scores by platform are presented in Table 3. The overall sample mean was 38.68 (SD = 10.16), with a median of 38.0 (range: 17–64). Traditional websites achieved the highest median score (Mdn = 48.5, IQR = 9.75), followed by YouTube (Mdn = 42.0, IQR = 9.00), TikTok (Mdn = 35.0, IQR = 8.00), and Instagram (Mdn = 27.5, IQR = 9.00). Using commonly adopted thresholds from subsequent DISCERN literature (16–26 = very poor, 27–38 = poor, 39–50 = fair, 51–62 = good, 63–80 = excellent), the modal category for websites was 'fair' (44%), for YouTube 'fair' (48%), for TikTok 'poor'

(55.3%), and for Instagram 'poor' (50%). Notably, 42% of Instagram items scored in the 'very poor' range, compared with 0% for both websites and YouTube.

Table 3. DISCERN Total Scores by Platform

Platform	n	M	SD	Mdn	Q1	Q3	Min	Max
Website	50	47.02	7.90	48.5	43.00	52.75	28	64
YouTube	50	42.88	7.10	42.0	38.00	47.00	30	57
TikTok	47	35.64	7.97	35.0	31.00	39.00	20	57
Instagram	50	28.98	6.82	27.5	25.00	34.00	17	52
Overall	197	38.68	10.16	38.0	—	—	17	64

Note. M = mean; SD = standard deviation; Mdn = median; Q1 = 25th percentile; Q3 = 75th percentile. DISCERN score categories are based on commonly adopted thresholds in the health information quality literature.

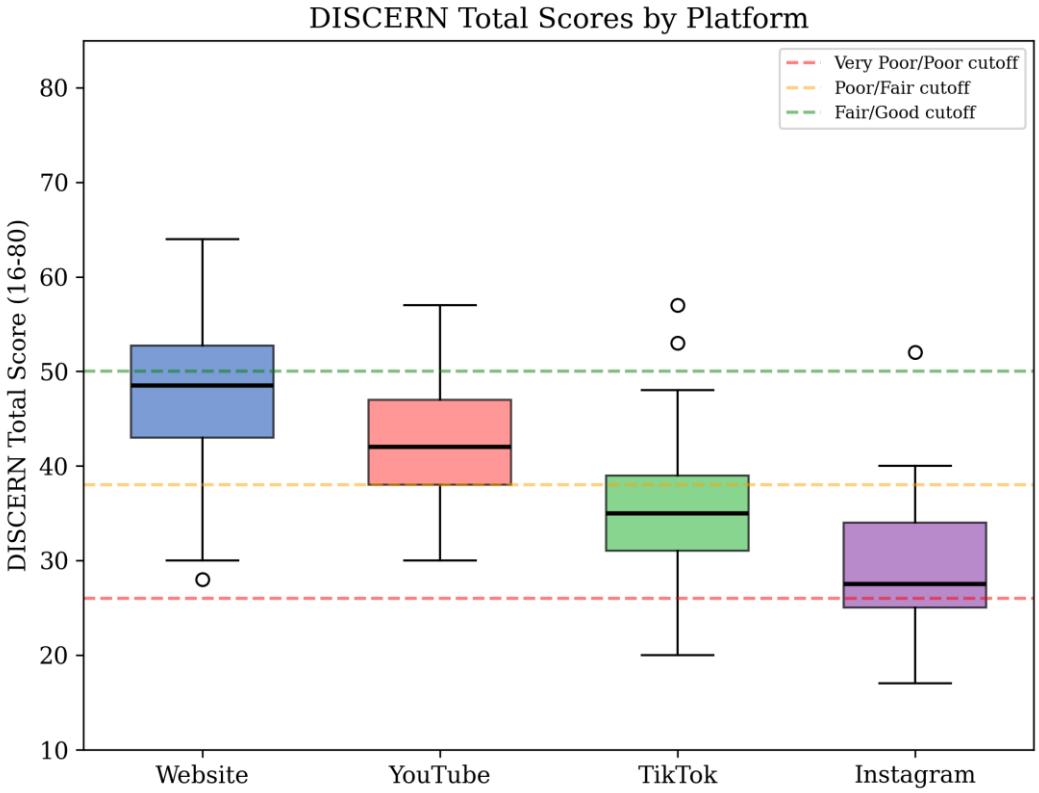


Figure 1. Box plot of DISCERN total scores by platform. The horizontal line within each box represents the median; box boundaries indicate Q1 and Q3; whiskers extend to 1.5 times the IQR.

A Kruskal-Wallis H test confirmed that DISCERN total scores differed significantly across the four platforms, $H(3) = 93.489$, $p < .001$. The estimated eta-squared (eta-squared = 0.469) indicated a large effect size. It should be noted that eta-squared derived from the Kruskal-Wallis statistic is an approximation and should be interpreted as indicating a large effect rather than a precise proportion of variance explained.

Pairwise Mann-Whitney U comparisons with Bonferroni correction revealed significant differences for all six platform pairs (Appendix Table C3). The largest effect sizes were observed for Website versus Instagram ($r_{rb} = -0.891$, large) and YouTube versus Instagram ($r_{rb} = -0.850$, large). The Website versus YouTube comparison, while statistically significant at the Bonferroni-adjusted threshold ($U = 1646$, $p = .038$, $r_{rb} = -0.316$), exhibited a small-to-medium effect size, suggesting that these two platforms are relatively more similar in information quality than other pairings. The remaining comparisons—Website versus TikTok ($U = 1972$, $p < .001$, $r_{rb} = -0.678$), YouTube versus TikTok ($U = 1775$, $p < .001$, $r_{rb} = -0.511$), and TikTok versus Instagram ($U = 1740$, $p < .001$, $r_{rb} = -0.480$)—all yielded medium to large effects.

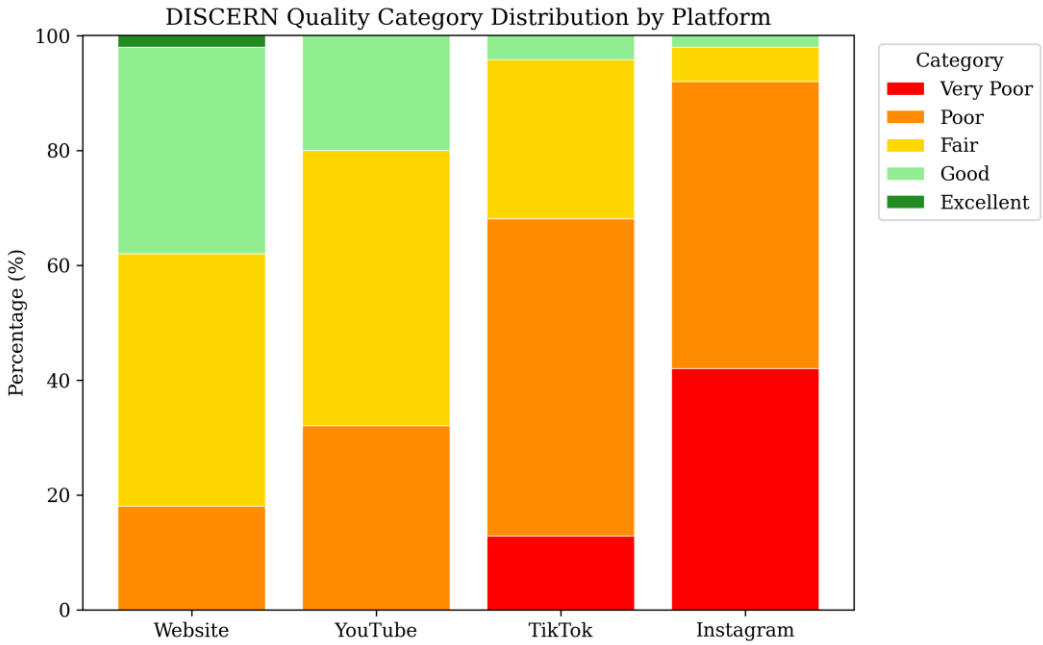


Figure 2. Distribution of DISCERN quality categories by platform. Categories are based on conventional thresholds: Very Poor (16–26), Poor (27–38), Fair (39–50), Good (51–62), and Excellent (63–80) (Sun et al., 2023).

DISCERN section scores followed a consistent pattern. For Section 1 (reliability of publication, Q1–Q8), median scores decreased from websites (Mdn = 25, IQR = 23–29) to YouTube (Mdn = 24, IQR = 21–27), TikTok (Mdn = 19, IQR = 17–22), and Instagram (Mdn = 15, IQR = 13–18). Section 2 (quality of treatment information, Q9–Q15) showed a parallel decline: websites (Mdn = 18, IQR = 16–21), YouTube (Mdn = 16, IQR = 14–18), TikTok (Mdn = 14, IQR = 11–16), and Instagram (Mdn = 10, IQR = 9–13).

3.4 JAMA Benchmark Compliance by Platform

JAMA benchmark compliance varied by criterion and platform (Table 4). A Kruskal-Wallis test on JAMA total scores (range 0–4) revealed no statistically significant difference across platforms, $H(3) = 5.123$, $p = .163$. However, this aggregate result is partly an artefact of the Currency criterion, which gives social media platforms an automatic advantage: posting dates are inherently visible (100% compliance) on social media but must be explicitly stated on websites (68% compliance). Criterion-level comparisons are therefore more informative than total JAMA scores in this study. As a sensitivity check, JAMA scores were also computed on a 0–3 scale excluding Currency. Under this recalculation, mean scores were: websites 0.76, YouTube 0.54, TikTok 0.43, and Instagram 0.20, reinforcing the transparency advantage of traditional websites when the platform-inherent date display is set aside.

Chi-square tests on individual JAMA criteria revealed significant platform differences for three of the four benchmarks. Attribution differed significantly across platforms ($\chi^2[3] = 24.255$, $p < .001$, Cramér's $V = 0.351$, medium effect), with websites (30%) and YouTube (16%) demonstrating the highest compliance, while TikTok (0%) provided no source citations whatsoever. Disclosure also differed significantly ($\chi^2[3] = 15.921$, $p = .001$, $V = 0.284$, medium effect), with websites (16%) and TikTok (12.8%) showing the highest rates of sponsorship declaration; YouTube and Instagram both recorded 0% disclosure compliance. Currency showed significant variation ($\chi^2[3] = 51.198$, $p < .001$, $V = 0.510$, large effect), driven by the contrast between traditional websites (68%) and social media platforms (100% each). Authorship did not differ significantly across platforms ($\chi^2[3] = 6.151$, $p = .105$, $V = 0.176$, small effect), with compliance ranging from 16% (Instagram) to 38% (YouTube).

Table 4. JAMA Benchmark Compliance by Platform (%)

Criterion	Website	YouTube	TikTok	Instagram
Authorship	30%	38%	29.8%	16%
Attribution	30%	16%	0%	4%
Disclosure	16%	0%	12.8%	0%
Currency	68%	100%	100%	100%
Mean Total (0–4)	1.44	1.54	1.43	1.20

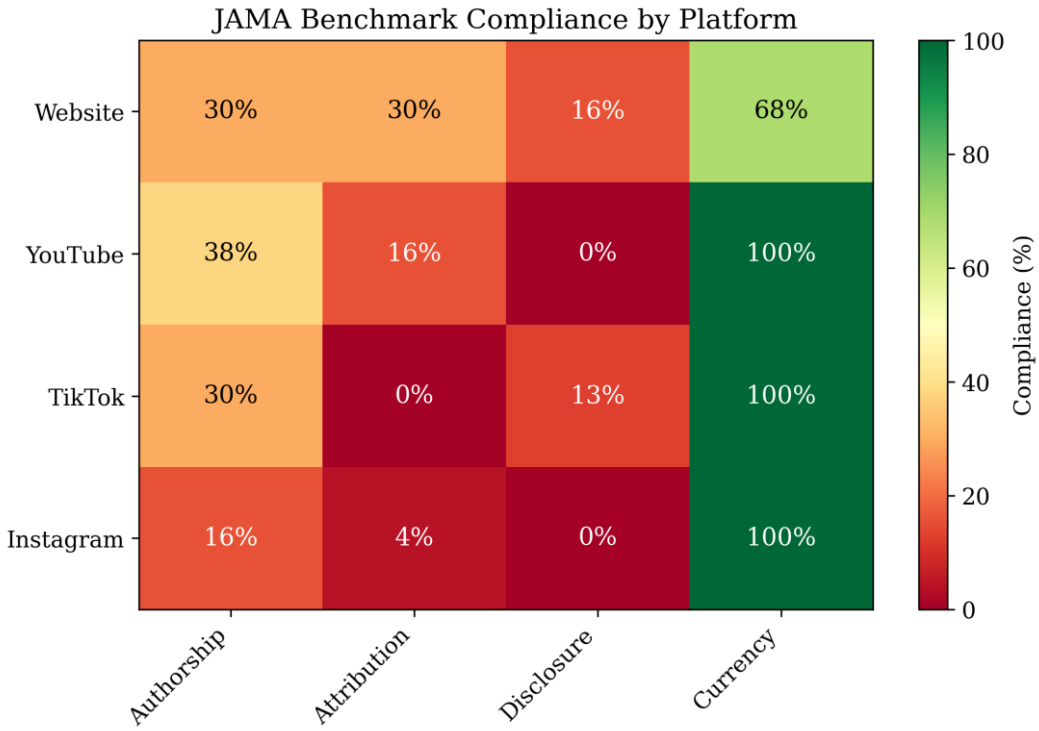


Figure 3. Heatmap of JAMA benchmark compliance across platforms. Darker cells indicate higher compliance rates.

3.5 Creator Type Analysis

To address Research Question 2, creators were initially dichotomised into professional (healthcare professionals and certified fitness professionals, n = 51) and non-professional (fitness influencers, general users, and organisations, n = 146) categories. A Mann-Whitney U test comparing DISCERN total scores between these two groups yielded U = 4457, p = .036, r_rb = −0.197, reaching statistical significance at the α = .05 level with a small effect size. Professional creators achieved a median DISCERN score of 41.0, compared with 37.0 for non-professional creators.

It is important to note that the binary classification is complicated by the treatment of organisations (n = 84, 42.6% of the sample). Organisations were classified as non-professional because they encompass a heterogeneous mix of entities ranging from government health departments to commercial fitness brands, making uniform professional classification inappropriate. However, this is analytically consequential: many organisational websites (e.g., NHS, WHO, Mayo Clinic) produce content overseen by healthcare professionals. As a sensitivity analysis, when organisations were excluded from the binary classification, the difference between professional (n = 51, Mdn = 41.0) and non-professional

creators (n = 62, comprising fitness influencers and general users, Mdn = 35.0) was significant (p = .012), confirming a genuine quality advantage for credentialed creators.

The five-category Kruskal-Wallis test provided a more granular picture, H(4) = 12.764, p = .012. Healthcare professionals scored highest (Mdn = 45.0), followed by certified fitness professionals (Mdn = 39.5), organisations (Mdn = 39.0), fitness influencers (Mdn = 36.0), and general users (Mdn = 33.0). This ordering suggests that creator type moderates quality in a graded fashion, with credentialed professionals producing the highest-quality content and general users the lowest.

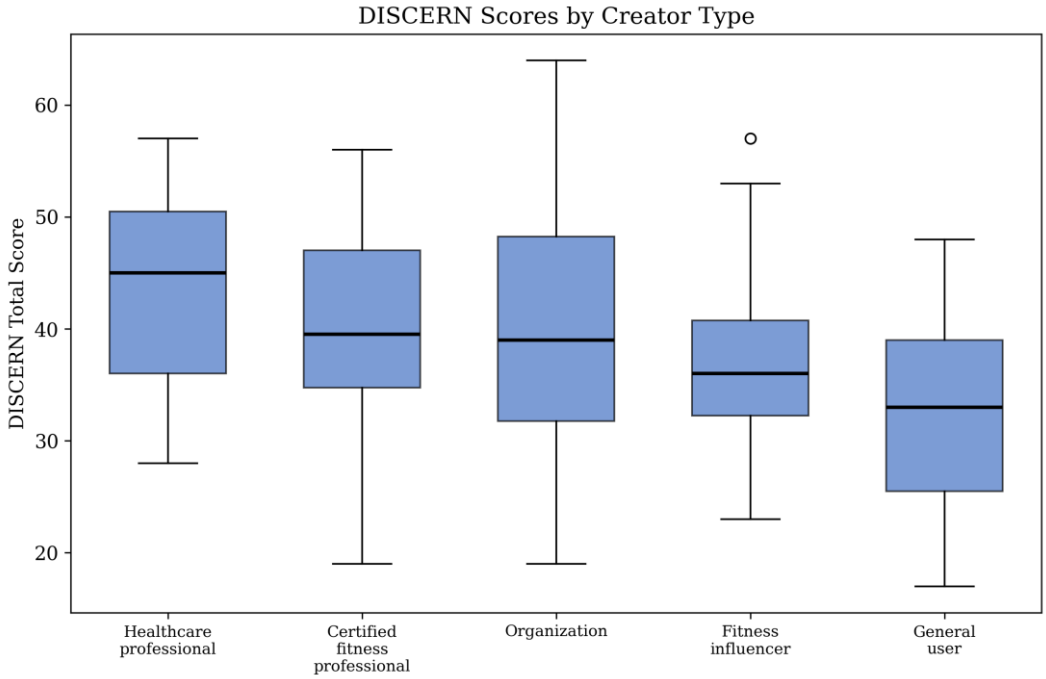


Figure 4. DISCERN total scores by creator type. Box plots show median, IQR, and individual observations.

3.6 Engagement-Quality Relationship

Research Question 3 examined whether engagement metrics were associated with information quality. Appendix Table C6 summarises the Spearman rank-order correlations between engagement indicators and DISCERN total scores.

At the overall sample level, views (available for YouTube and TikTok, n = 97) showed a negligible and non-significant correlation with DISCERN scores (rho = -0.044, p = .669). Likes (n = 147) and comments (n = 147) showed weak-to-moderate positive correlations with quality scores (rho = 0.284, p < .001 and rho = 0.371, p < .001, respectively). Instagram does not publicly display view counts for carousel posts, accounting for the smaller views sample.

These overall positive correlations for likes and comments must be interpreted with considerable caution, as they represent a largely confounded association driven by between-social-platform differences rather than by a consistent within-platform relationship. When examined within individual platforms, TikTok (rho = 0.109, p = .467) and Instagram (rho = -0.174, p = .226) showed non-significant likes–quality correlations. However, YouTube demonstrated a significant negative likes–quality correlation (rho = -0.286, p = .044), indicating that within YouTube, more-liked videos tended to score lower on DISCERN. This platform-specific finding is an exception to the broader Simpson's paradox pattern and suggests that the overall positive association between likes and quality is predominantly an ecological artefact rather than reflecting a genuine within-platform relationship. The overall positive correlation is an ecological artefact of social-platform differences rather than evidence that higher-quality content attracts more engagement within platforms.

For views, a significant negative correlation was observed within YouTube (rho = -0.340, p = .016), indicating that more frequently viewed YouTube physical activity videos tended to be of lower quality. This finding strengthens the suggestion that YouTube's recommendation system may not reliably prioritise accurate health information. Within TikTok, the views–quality correlation was negligible and non-significant (rho = -0.057, p = .702).

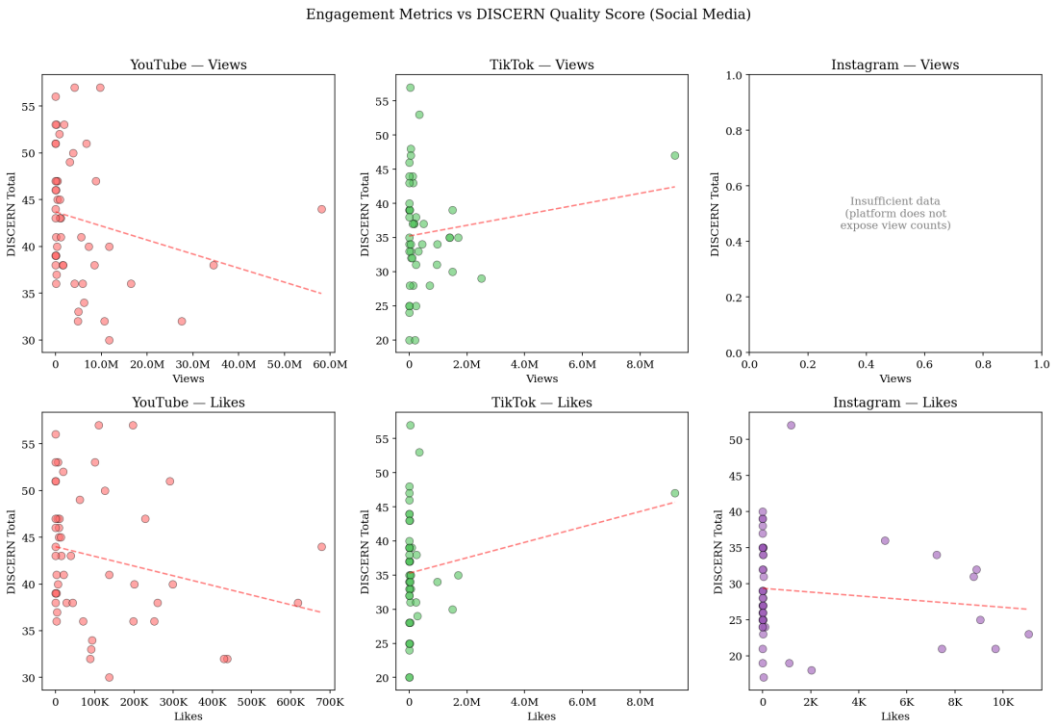


Figure 5. Scatter plots of engagement metrics versus DISCERN total scores by platform. Top row: views (Instagram lacks public view counts for non-Reel posts). Bottom row: likes. Trend lines are fitted using linear regression.

3.7 DISCERN Question-Level Patterns

Figure 6 presents a heatmap of mean scores for each of the 16 DISCERN questions across platforms. Several patterns merit attention. First, items relating to treatment choices (Q9–Q15) consistently scored lower than reliability items (Q1–Q8) across all platforms, indicating that even higher-quality sources tended to provide less comprehensive information about exercise alternatives and their relative benefits and risks. Second, Q4 ('Is it clear what sources of information were used to compile the publication?') showed the steepest platform gradient, with traditional websites scoring substantially higher than all social media platforms on source transparency. Q5 ('Is it clear when the information used or reported was produced?') showed a different pattern: YouTube scored highest on this currency item because upload dates are automatically and prominently displayed, whereas Instagram and TikTok scored lowest despite also auto-displaying post dates, likely because their content less frequently referenced when the underlying information was produced. Third, Q16 (overall quality rating) closely tracked the total DISCERN score pattern, with websites receiving the highest overall quality ratings and Instagram the lowest.

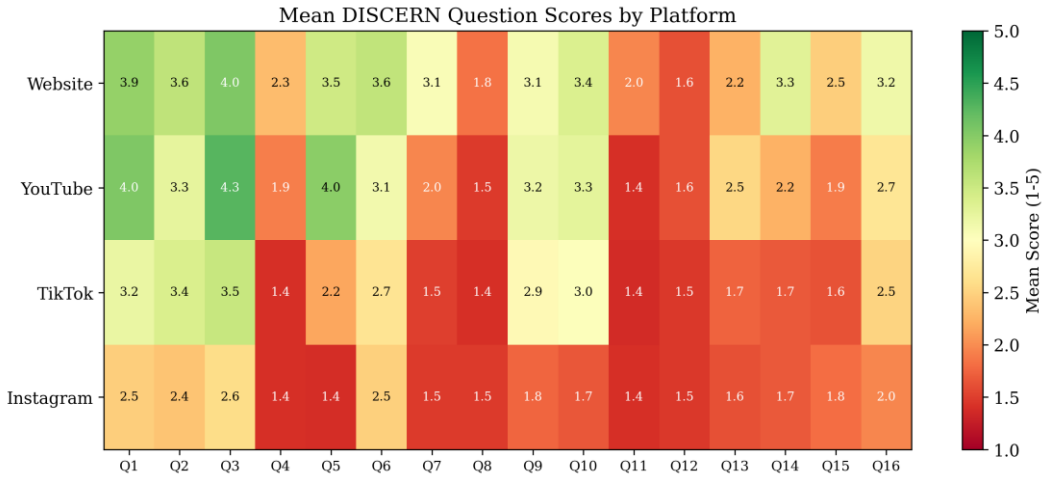


Figure 6. Heatmap of mean DISCERN item scores (Q1–Q16) by platform. Darker shading indicates higher mean scores (maximum = 5).

3.8 Content Format and Search Term Effects

Because content format and platform are perfectly confounded in this dataset (each platform uses a single exclusive format), DISCERN score differences by format exactly mirror the platform differences reported in section 3.3. Text articles (n = 50, websites) achieved the highest median score (Mdn = 48.5, M = 47.0). Long-form videos exceeding five minutes (n = 50, YouTube) scored second highest (Mdn = 42.0, M = 42.9). Short videos under 60 seconds

(n = 47, TikTok) scored lower (Mdn = 35.0, M = 35.6). Carousel posts (n = 50, Instagram) scored lowest (Mdn = 27.5, M = 29.0). Because format and platform cannot be disentangled, attributing these quality differences to format per se—rather than to other platform-level factors such as user norms, algorithmic incentives, and creator demographics—is not warranted from this dataset alone.

Search terms also influenced quality scores. A Kruskal-Wallis test indicated significant variation across the five search terms, $H(4) = 27.255$, $p < .001$ (see Appendix Table C10 for descriptive statistics by search term). Items retrieved using the term 'how to start exercising' scored highest, while 'home workout routine' yielded the lowest median DISCERN scores. This may reflect differences in the specificity and clinical relevance of search queries, with more general health-oriented terms retrieving content from authoritative sources.

CHAPTER 4. CONSIDERATIONS

4.1 Summary of Key Findings

This cross-sectional content analysis evaluated the quality of physical activity information across four platform types using the DISCERN instrument and JAMA benchmarks. The results reveal a clear and statistically significant quality gradient: traditional websites scored highest (Mdn = 48.5), followed by YouTube (Mdn = 42.0), TikTok (Mdn = 35.0), and Instagram (Mdn = 27.5). The Kruskal-Wallis test confirmed significant differences across platforms, $H(3) = 93.489$, $p < .001$, with a large eta-squared approximation (0.469). All six pairwise comparisons were statistically significant, although the Website versus YouTube contrast yielded a notably smaller effect ($p = .038$, $r_{rb} = -0.316$) compared with the other five pairs (all $p < .001$).

JAMA benchmark compliance did not differ significantly across platforms when aggregated ($H[3] = 5.123$, $p = .163$), though significant platform differences emerged for individual criteria—Attribution, Disclosure, and Currency—but not Authorship. Creator type exerted a significant effect on quality in the binary analysis ($p = .036$), and the five-category Kruskal-Wallis test was also significant ($p = .012$). Engagement–quality correlations were largely non-significant within platforms, except for YouTube, where both views and likes showed significant negative correlations with quality; the overall positive correlation between likes and quality across social-media items was identified as primarily a Simpson's paradox artefact of between-social-platform differences.

4.2 Comparison with Existing Literature

The finding that traditional websites outperform social media platforms in health information quality is consistent with a substantial body of prior research. The 2025 JMIR meta-analysis reported a pooled percentage-standardised DISCERN score of 43.58% (95% CI [37.80%, 49.35%]) across studies evaluating cancer-related information on new media (Liu et al., 2025). Direct numerical comparison with the present study's raw DISCERN totals (scored out of 80) is not appropriate because the meta-analysis synthesised percentage-standardised scores across heterogeneous studies. Even so, the present findings are directionally consistent with that literature, particularly in showing lower quality for highly visual or video-led social media formats than for websites and longer-form content.

The Instagram findings are particularly noteworthy. With 42% of items falling in the 'very poor' category and a median DISCERN score of 27.5, Instagram emerged as the lowest-quality platform by a substantial margin. This aligns with the broader pattern described by Li et al. (2024), who found that most TikTok health-related short videos were of moderate to low quality and that medically credentialed creators typically represented only a minority of content producers, and with Haghighi and Farhadloo (2025), who reported that 81% of Twitter health information was of low quality, with 95% of items meeting two or fewer JAMA criteria. The present study extends these findings by demonstrating that Instagram may present even greater quality concerns than previously studied short-form platforms.

The relatively modest difference between websites and YouTube (median difference of 6.5 points, $p = .038$) is an important nuance. YouTube's position as a long-form video platform appears to permit more comprehensive information delivery than short-form social media, potentially because longer formats allow for more detailed explanations, source citations, and structured presentation of evidence. This finding is consistent with more recent review evidence showing that longer-form health videos tend to permit greater informational completeness than short-form social-media content (Osman et al., 2022; Rodriguez-Rodriguez et al., 2022).

The non-significant JAMA aggregate result ($p = .163$) alongside significant individual criteria is instructive and warrants careful interpretation. Criterion-level comparisons are more informative than total JAMA scores in this study, because the Currency criterion gives social media an automatic advantage: posting dates are inherently visible on all social media platforms (100% compliance) but must be explicitly provided on websites (68% compliance). This means total JAMA scores partly reflect platform interface design rather than editorial accountability. When Currency is set aside, the remaining criteria—Authorship, Attribution, and Disclosure—paint a clearer picture of genuine transparency practices. Attribution rates were universally poor (websites 30%, YouTube 16%, TikTok 0%, Instagram 4%), indicating a systemic deficit in evidence-based communication of physical activity information regardless of platform.

The significant variation in quality across search terms ($H[4] = 27.255$, $p < .001$) is a secondary finding that merits attention. The query ‘how to start exercising’ retrieved the highest-quality content, while ‘home workout routine’ retrieved the lowest-quality content. This likely reflects differences in the clinical relevance and specificity of queries: more general health-oriented terms may retrieve content from authoritative institutional sources,

whereas commercially oriented terms such as ‘home workout routine’ may disproportionately surface influencer-generated content. This finding suggests that the search query itself functions as an implicit filter on information quality, with practical implications for how public health practitioners frame their recommendations to information-seeking audiences.

4.3 Implications for Public Health Practice

These findings carry several implications for public health practice. First, the pronounced quality gradient across platforms suggests that public health agencies should prioritise maintaining high-quality website content as an anchor for physical activity information, while simultaneously investing in platform-native content for social media. Simply repurposing website content for Instagram or TikTok is unlikely to be effective; instead, organisations should develop format-specific strategies that maximise information quality within the constraints of each platform.

Second, the finding that engagement metrics largely do not predict quality within platforms undermines the common assumption that popular content is reliable content. This has implications for health literacy education: consumers should be taught that high view counts and engagement do not signal information trustworthiness. Notably, within YouTube, both views and likes showed significant negative correlations with quality (views: rho = -0.340, p = .016; likes: rho = -0.286, p = .044), suggesting that the platform’s recommendation system may actively promote lower-quality physical activity content. This is a concerning finding that warrants dedicated algorithmic audit research, though the cross-sectional design cannot establish causality.

Third, the creator type analysis reveals that both the binary and five-category comparisons reached significance (p = .036 and p = .012, respectively). The five-category analysis provides the more granular picture: healthcare professionals score highest (Mdn = 45.0), followed by certified fitness professionals (Mdn = 39.5), organisations (Mdn = 39.0), fitness influencers (Mdn = 36.0), and general users lowest (Mdn = 33.0). This pattern suggests that creator type moderates quality in a graded fashion. The binary result, while significant, is somewhat diluted by grouping organisations (n = 84)—which include high-quality institutional sources such as NHS and WHO—into the non-professional category.

4.4 Strengths of the Study

This study possesses several methodological strengths. To our knowledge, it is among the first investigations to directly compare physical activity information quality across four distinct platform types—traditional websites, YouTube, TikTok, and Instagram—using both the DISCERN instrument and JAMA benchmarks within a single study. This multi-platform, dual-instrument approach provides a more comprehensive assessment than studies examining single platforms or single quality tools.

The study employed a systematic sampling strategy with five distinct search terms representing common physical activity queries, enhancing ecological validity. The inter-rater reliability was excellent for DISCERN (ICC = 0.932), providing confidence in the scoring process. The sample of 197 items across four platforms is larger than many comparable studies, affording sufficient statistical power for the planned analyses. Furthermore, the operationalisation of DISCERN and JAMA criteria for social media contexts was documented transparently, enabling replication.

4.5 Limitations

Several limitations must be acknowledged. First, the cross-sectional design captures a single snapshot of online content, which may not reflect temporal trends in information quality. Platform algorithms, content policies, and user behaviour evolve rapidly, and findings may not generalise to other time points.

Second, the study was restricted to English-language content retrieved from a single geographic location (Lithuania). Search results are personalised by location, language, and browsing history, meaning that users in other regions may encounter different content for identical search terms.

Third, and critically, scoring of video and image-based content was restricted to information visible or audible to a typical user at the point of access, including spoken narration, on-screen text, captions, and profile metadata, rather than full transcription or frame-by-frame coding. This approach may still miss subtle cues or evidence embedded in longer viewing sessions. It is therefore important to acknowledge that the apparent quality gradient could partially reflect methodological differences in how text-based and audiovisual content were assessed rather than true differences in information quality.

Fourth, data collection employed a mixed approach: Playwright automation was used for traditional websites and YouTube, while TikTok and Instagram content was collected manually due to platform restrictions on automated scraping. This introduces potential selection bias, as the manual collection process for TikTok and Instagram may have been influenced by factors not present in the automated approach, despite adherence to the same top-10 sampling protocol.

Fifth, the top-10 sampling strategy, while ecologically valid (as most users do not scroll beyond the first page of results), captures only a fraction of available content. Results may differ for content appearing in later search positions. Additionally, the DISCERN instrument was originally developed for evaluating written treatment information for patients (Charnock et al., 1999) and has been adapted here beyond its original scope to evaluate physical activity content on social media. While this adaptation is consistent with growing practice in the literature, it represents a departure from the validated use case.

Sixth, engagement metrics were not uniformly available across platforms. Traditional websites do not provide standardised engagement data (likes, shares), and the metrics that were available may not be comparable across platforms with different user bases and interaction paradigms.

Seventh, the classification of organisations (n = 84, 42.6% of the sample) as 'non-professional' in the binary creator type analysis is analytically consequential. Many organisations (e.g., NHS, WHO, university hospitals) produce content overseen by healthcare professionals, meaning this classification likely diluted the professional category's apparent quality advantage. Future studies should consider more granular organisation subtyping.

Eighth, the DISCERN quality categories used in this study (Very Poor 16–26, Poor 27–38, Fair 39–50, Good 51–62, Excellent 63–80) are commonly adopted thresholds in the health information quality literature but were not specified in the original DISCERN validation by Charnock et al. (1999); one recent example using the same cut-points is Sun et al. (2023). These cut-points have been adopted from subsequent studies and should be treated as conventional rather than formally validated boundaries.

4.6 Recommendations for Future Research

Several directions for future research emerge from this study. First, longitudinal designs could track how physical activity information quality changes over time as platforms modify their algorithms and content policies. Second, multilingual studies incorporating non-English

content would address a significant gap, given that the majority of global internet users access health information in languages other than English. Third, future studies should incorporate full transcription and more fine-grained coding of audio-visual content to extend the point-of-access scoring approach used here and determine whether subtle spoken or visual cues alter quality ratings. Fourth, the role of platform algorithms in promoting or suppressing high-quality health content warrants dedicated investigation, particularly given the significant negative views–quality and likes–quality correlations observed on YouTube. Fifth, intervention studies could evaluate whether platform-specific quality labels or health information verification badges improve the quality of content that users encounter and trust. Finally, qualitative research exploring how users perceive and evaluate physical activity information across different platforms would complement the quantitative quality assessment approach used here.

CONCLUSIONS

This study set out to evaluate the quality of physical activity information across traditional websites and social media platforms using the DISCERN instrument and JAMA benchmarks. Three research questions guided the investigation. The conclusions for each are presented below.

Research Question 1: How does physical activity information quality differ between traditional websites and social media platforms? The results demonstrate a clear, statistically significant quality gradient across platforms, $H(3) = 93.489$, $p < .001$. Traditional websites achieved the highest median DISCERN score (Mdn = 48.5), followed by YouTube (Mdn = 42.0), TikTok (Mdn = 35.0), and Instagram (Mdn = 27.5). All six pairwise comparisons were statistically significant, though the website–YouTube difference was more modest ($p = .038$) than the remaining five comparisons (all $p < .001$). JAMA benchmark compliance did not differ significantly across platforms in aggregate ($p = .163$), although individual criteria showed significant platform variation for Attribution, Disclosure, and Currency. It can be concluded that a substantial quality gap exists between traditional websites and short-form social media platforms, with Instagram representing the most concerning source of physical activity information.

Research Question 2: Does creator type moderate quality scores? The binary professional versus non-professional comparison was statistically significant ($U = 4457$, $p = .036$, $r_{rb} = -0.197$), and the five-category creator type analysis confirmed this ($H[4] = 12.764$, $p = .012$). Healthcare professionals scored highest (Mdn = 45.0), followed by certified fitness professionals (Mdn = 39.5), organisations (Mdn = 39.0), fitness influencers (Mdn = 36.0), and general users lowest (Mdn = 33.0). It can be concluded that creator type influences information quality in a graded fashion, with credentialed professionals producing measurably higher-quality content.

Research Question 3: What is the relationship between engagement metrics and information quality? Overall correlations suggested weak-to-moderate positive associations between likes (rho = 0.284, $p < .001$), comments (rho = 0.371, $p < .001$), and DISCERN scores; however, these aggregate associations are primarily an ecological artefact of between-social-platform differences (Simpson's paradox). Within-platform analyses revealed that TikTok and Instagram showed no significant likes–quality correlations. Critically, YouTube demonstrated significant negative correlations for both views (rho = -0.340, $p = .016$) and likes (rho = -

0.286, $p = .044$) with quality, indicating that more popular YouTube content tended to score lower on DISCERN. Views showed a negligible overall correlation (rho = -0.044, $p = .669$). It can be concluded that engagement metrics do not appear to reliably indicate information quality and that popular content is not necessarily accurate content.

Additionally, search query phrasing significantly influenced the quality of content retrieved ($H[4] = 27.255$, $p < .001$), with 'how to start exercising' yielding the highest-scoring content and 'home workout routine' the lowest. This suggests that the wording of health information queries functions as an implicit quality filter, a finding with practical implications for how clinicians advise patients on information seeking.

RECOMMENDATIONS

1. Public health organisations should maintain high-quality traditional websites as anchor resources for physical activity information and should also produce platform-native short-form content that clearly states aims, cites sources, and acknowledges uncertainty.

2. Healthcare professionals should engage more actively on social media, display their credentials prominently, and include source citations wherever feasible to counter the dominance of unqualified creators.

3. Platform developers and policy makers should strengthen transparency standards for health-related content, including prompts for evidence citation, clearer credential signalling, and greater scrutiny of recommendation systems that amplify highly viewed but lower-quality content.

4. Health literacy initiatives should explicitly teach that views, likes, and shares are not reliable indicators of information quality and should train users to check author credentials and cited sources.

5. Consumers seeking physical activity guidance should use recognised health organisation websites as their primary source and should verify creator credentials and cross-check claims against authoritative guidelines when using social media.

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ANNEXES

APPENDIX A: DISCERN INSTRUMENT

The DISCERN instrument (Charnock et al., 1999) is a validated tool for assessing the quality of written consumer health information. It consists of 16 questions across three sections, each scored on a 5-point Likert scale (1 = Definitely No to 5 = Definitely Yes). The instrument was originally designed for patient information leaflets but has been widely adapted for online health content assessment.

Section 1: Reliability of the Publication (Questions 1–8)

Q1: Are the aims clear?

Does the publication clearly state what it is about and what topics will be covered?

Q2: Does it achieve its aims?

Does the content deliver on the stated purpose?

Q3: Is it relevant?

Is the information relevant to the target audience?

Q4: Is it clear what sources of information were used?

Are references, sources, or evidence cited?

Q5: Is it clear when the information was produced?

Is the date of creation or last update stated?

Q6: Is it balanced and unbiased?

Does the content present information objectively, or is it one-sided?

Q7: Does it provide details of additional sources of support?

Are links to further reading, helplines, or professional resources provided?

Q8: Does it refer to areas of uncertainty?

Does the content acknowledge limitations, uncertainties, or areas where evidence is lacking?

Section 2: Quality of Information on Treatment Choices (Questions 9–15)

Q9: Does it describe how each treatment works?

Does the content explain the mechanism or rationale behind recommended exercises or activities?

Q10: Does it describe the benefits of each treatment?

Are the expected benefits of the recommended physical activity clearly stated?

Q11: Does it describe the risks of each treatment?

Are potential risks, contraindications, or side effects of the recommended exercises discussed?

Q12: Does it describe what would happen if no treatment is used?

Does the content explain the consequences of physical inactivity?

Q13: Does it describe how treatment choices affect overall quality of life?

Does the content address how physical activity affects wellbeing beyond the specific condition?

Q14: Is it clear that there may be more than one possible treatment choice?

Does the content present multiple exercise options rather than a single approach?

Q15: Does it provide support for shared decision-making?

Does the content encourage consulting a professional or making personalised decisions?

Section 3: Overall Quality Rating (Question 16)

Q16: Based on the answers to all of the above questions, rate the overall quality of the publication as a source of information about treatment choices.

Scoring and Interpretation

Each question is scored from 1 (Definitely No) to 5 (Definitely Yes). The total score ranges from 16 to 80. The following categories are used for interpretation:

Score Range	Category	Interpretation
16–26	Very Poor	Serious or extensive shortcomings
27–38	Poor	Potentially important but not serious shortcomings
39–50	Fair	Some shortcomings but not serious
51–62	Good	Minimal shortcomings
63–80	Excellent	Minimal shortcomings with additional features

APPENDIX B: JAMA BENCHMARKS — OPERATIONALISATION FOR SOCIAL MEDIA

The JAMA benchmarks (Silberg et al., 1997) were originally proposed as minimum standards for health information published on the World Wide Web. They comprise four accountability criteria, each assessed as present (Y) or absent (N). This appendix details how each criterion was operationalised for the four platform types assessed in this study.

1. Authorship

Definition: Authors and contributors should be identified, along with their affiliations and relevant credentials.

Platform	Operationalisation
Traditional websites	Author name and credentials (e.g., MD, PhD, CSCS) stated on the page, in a byline, or in an About the Author section.
YouTube	Creator credentials stated in the video description, channel About page, or verbally in the video introduction.
TikTok	Creator bio states professional credentials (e.g., Certified Personal Trainer, Physiotherapist) OR account is verified as a professional.
Instagram	Creator bio states professional credentials OR profile is a verified professional/business account with stated qualifications.

2. Attribution

Definition: References and sources for all content should be listed clearly.

Platform	Operationalisation
Traditional websites	In-text citations, a reference list, bibliography, or hyperlinks to source materials are provided.
YouTube	Sources cited in the video description, shown on screen, or verbally referenced with sufficient detail to locate.
TikTok	Sources cited in the caption, shown as on-screen text, or referenced verbally (e.g., According to the WHO guidelines).
Instagram	Sources cited in the post caption, shown on carousel slides, or included as tagged references.

3. Disclosure

Definition: Website ownership, sponsorship, advertising, underwriting, commercial funding, or potential conflicts of interest should be disclosed.

Platform	Operationalisation
Traditional websites	A disclosure statement, privacy policy identifying sponsorships, or declaration of conflicts of interest is present.
YouTube	Video includes a paid promotion disclosure, sponsorship mention, or the creator states potential conflicts of interest.
TikTok	Post includes #ad, #sponsored, Paid partnership label, or the creator verbally discloses commercial relationships.

Instagram	Post includes Paid partnership with label, #ad, #sponsored tags, or disclosure in the caption.

4. Currency

Definition: Dates of content posting and any updates should be clearly displayed.

Platform	Operationalisation
Traditional websites	Publication date or last updated date is visible on the page.
YouTube	Upload date is displayed by default on the platform (always Y for YouTube).
TikTok	Post date is displayed by default on the platform (always Y for TikTok).
Instagram	Post date is displayed by default on the platform (always Y for Instagram).

Scoring

Each criterion is scored as Y (Yes, criterion met) or N (No, criterion not met). The JAMA Total score is the count of criteria met, ranging from 0 to 4.

Appendix C: Data Summary Tables

Table C1. DISCERN Total Score Descriptive Statistics by Platform

Platform	n	M	SD	Mdn	Q1	Q3	Min	Max
Website	50	47.02	7.90	48.5	43.00	52.75	28	64
YouTube	50	42.88	7.10	42.0	38.00	47.00	30	57
TikTok	47	35.64	7.97	35.0	31.00	39.00	20	57
Instagram	50	28.98	6.82	27.5	25.00	34.00	17	52
Overall	197	38.68	10.16	38.0	—	—	17	64

Table C2. DISCERN Quality Categories by Platform, n (%)

Platform	Very Poor	Poor	Fair	Good	Excellent
Website	0 (0%)	9 (18%)	22 (44%)	18 (36%)	1 (2%)
YouTube	0 (0%)	16 (32%)	24 (48%)	10 (20%)	0 (0%)
TikTok	6 (12.8%)	26 (55.3%)	13 (27.7%)	2 (4.3%)	0 (0%)
Instagram	21 (42%)	25 (50%)	3 (6%)	1 (2%)	0 (0%)

Note. Categories: Very Poor (16–26), Poor (27–38), Fair (39–50), Good (51–62), Excellent (63–80) (Sun et al., 2023).

Table C3. Pairwise Platform Comparisons (Mann-Whitney U, Bonferroni-Adjusted)

Comparison	U	p (adjusted)	r_rb
Website vs YouTube	1646	.038	-0.316
Website vs TikTok	1972	< .001	-0.678
Website vs Instagram	2364	< .001	-0.891
YouTube vs TikTok	1775	< .001	-0.511
YouTube vs Instagram	2313	< .001	-0.850
TikTok vs Instagram	1740	< .001	-0.480

Table C4. JAMA Benchmark Compliance by Platform (%)

Criterion	Website	YouTube	TikTok	Instagram
Authorship	30%	38%	29.8%	16%
Attribution	30%	16%	0%	4%
Disclosure	16%	0%	12.8%	0%
Currency	68%	100%	100%	100%

Table C5. Inter-Rater Reliability Statistics

Measure	Statistic	Interpretation
DISCERN Total (ICC[3,1])	0.932 (95% CI [0.870, 0.960])	Excellent
JAMA Authorship (kappa)	0.719	Substantial
JAMA Attribution (kappa)	0.686	Substantial
JAMA Disclosure (kappa)	0.305	Fair (kappa paradox)
JAMA Currency (kappa)	0.215	Fair (kappa paradox)

Table C6. Spearman Correlations Between Engagement Metrics and DISCERN Scores

Metric / Platform	rho	p	Interpretation
Views (overall, n=97)	-0.044	.669	Negligible, ns
Likes (overall, n=147)	0.284	< .001	Weak positive (confounded)
Comments (overall, n=147)	0.371	< .001	Moderate positive (confounded)
YouTube views	-0.340	.016	Significant negative
YouTube likes	-0.286	.044	Significant negative
TikTok views	-0.057	.702	ns
TikTok likes	0.109	.467	ns
Instagram likes	-0.174	.226	ns
Note			Overall

			likes/comments confounded; YouTube shows real negative correlations

Table C7. DISCERN Section Scores by Platform (Median, IQR)

Platform	Section 1 Mdn	Section 1 IQR	Section 2 Mdn	Section 2 IQR
Website	25	23–29	18	16–21
YouTube	24	21–27	16	14–18
TikTok	19	17–22	14	11–16
Instagram	15	13–18	10	9–13

Table C8. DISCERN Scores by Creator Type

Creator Type	n	Mdn DISCERN
Healthcare Professional	19	45.0
Certified Fitness Professional	32	39.5
Fitness Influencer	38	36.0
General User	24	33.0
Organisation	84	39.0

Table C9. DISCERN Scores by Content Format (= Platform)

Format	Platform	n	Mdn	M
Text Article	Website	50	48.5	47.0
Long Video (>5 min)	YouTube	50	42.0	42.9
Short Video (<60 s)	TikTok	47	35.0	35.6
Carousel	Instagram	50	27.5	29.0

Table C10. DISCERN Total Scores by Search Term

Search Term	n	Mdn	Q1	Q3	M	SD
How to start exercising	40	46.0	37.75	51.25	43.83	9.15
Best exercises to lose weight	40	37.0	28.75	45.00	36.70	9.47
Strength training for beginners	37	39.0	33.00	43.00	39.62	8.86
Physical activity guidelines	40	39.5	32.00	51.00	40.70	11.23
Home workout routine	40	34.0	25.00	38.00	32.60	8.52

Note. Kruskal-Wallis $H(4) = 27.255$, $p < .001$. n values reflect the final audited sample ($N = 197$); exact counts per search term vary slightly due to TikTok audit exclusions.

APPENDIX D: STATISTICAL ANALYSIS CODE

The complete statistical analysis was conducted using Python 3.13 with the following packages: pandas, scipy, pingouin, matplotlib, and seaborn. The full analysis script (run_analysis.py) is available in the project repository. Key analysis functions are summarised below.

D.1 Software Environment

Python 3.13

pandas 2.2.x — data manipulation

scipy 1.14.x — Kruskal-Wallis H, Mann-Whitney U, Chi-square, Spearman correlation

pingouin 0.5.x — Intraclass Correlation Coefficient (ICC)

matplotlib 3.9.x + seaborn 0.13.x — figure generation

D.2 Key Statistical Tests

Inter-rater reliability. ICC(3,1) computed via pingouin.intraclass_corr() for DISCERN total scores. Cohen's kappa calculated manually for binary JAMA criteria using observed vs. expected agreement.

Platform comparisons (DISCERN). Kruskal-Wallis H test (scipy.stats.kruskal) followed by Mann–Whitney U pairwise comparisons with Bonferroni correction. The Kruskal-Wallis omnibus effect size was computed as $(H - k + 1) / (N - k)$, where H is the Kruskal-Wallis statistic, k the number of groups, and N the total sample size (Tomczak & Tomczak, 2014). Pairwise effect sizes were reported as rank-biserial correlation r.

Platform comparisons (JAMA). Chi-square test of independence (scipy.stats.chi2_contingency) for each JAMA criterion. Fisher's exact test used when expected cell counts < 5. Effect size reported as Cramér's V.

Creator type analysis. Mann-Whitney U test (scipy.stats.mannwhitneyu) comparing professional vs. non-professional creators. Effect size: rank-biserial correlation.

Engagement-quality correlations. Spearman's rank correlation (scipy.stats.spearmanr) between engagement metrics and DISCERN total, both overall and stratified by platform.

D.3 Reproducibility

All data, scripts, and analysis outputs are maintained in a version-controlled Git repository available at https://github.com/ayothedoc3/masters-thesis (accessed 26 March 2026). The repository snapshot current at the time of submission contains the materials supporting the submitted thesis. The complete analysis can be reproduced by running: python analysis/run_analysis.py from the project root directory. The final audited input files are data/csv/master_dataset_corrected.csv and data/csv/second_rater_scores.csv. The final audited analytical sample contains 197 items: 50 websites, 50 YouTube videos, 47 TikTok videos, and 50 Instagram posts. The script normalises common content-format label variants before analysis and writes all output files (figures, tables, and analysis_report.txt) to analysis/output/.

APPENDIX E: DRAFT SCIENTIFIC ARTICLE PREPARED FOR JOURNAL SUBMISSION

Target journal format: JMIR Public Health and Surveillance (manuscript-style draft).

Quality of Physical Activity Information on Traditional Websites Versus Social Media Platforms: A Cross-Sectional Content Analysis Using DISCERN and JAMA Benchmarks

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Abstract

Background: Online physical activity information is increasingly consumed through both traditional websites and social media platforms, yet direct cross-platform comparisons remain limited. Objective: To compare the quality of physical activity information on traditional websites, YouTube, TikTok, and Instagram using DISCERN and JAMA benchmarks. Methods: A cross-sectional content analysis was conducted on 197 publicly accessible English-language items retrieved with five standardized search terms. Quality was assessed with the 16-item DISCERN instrument and four JAMA accountability criteria. Inter-rater reliability was evaluated on 40 items. Results: Quality differed significantly across platforms, with websites performing best, followed by YouTube, TikTok, and Instagram. JAMA compliance was low across all platforms. Creator type was associated with quality, and YouTube views and likes showed negative correlations with DISCERN scores. Conclusions: Traditional websites remain the most reliable source of physical activity information, whereas short-form social media content demonstrates weaker quality and transparency.

Keywords: physical activity; information quality; social media; DISCERN; JAMA

Introduction

The public now accesses physical activity guidance through a fragmented digital ecosystem in which institutional websites compete with user-generated and algorithmically promoted social media content. Although this improves reach, it also raises concerns about the credibility, transparency, and safety of the information users encounter. Prior studies have often examined one platform at a time, making it difficult to compare the quality of

information across major online environments. This study addressed that gap by evaluating four platform types under a common protocol.

Methods

A cross-sectional content analysis was performed using five standardized search terms: how to start exercising, best exercises to lose weight, strength training for beginners, physical activity guidelines, and home workout routine. The final audited sample comprised 197 items: 50 traditional websites, 50 YouTube videos, 47 TikTok posts, and 50 Instagram posts. Information quality was rated with the DISCERN instrument and the four JAMA criteria of authorship, attribution, disclosure, and currency. A second rater independently scored 40 items to assess reliability. Non-parametric tests were used because the main outcome measures were ordinal and non-normally distributed.

Results

A clear quality gradient emerged across platforms. Traditional websites achieved the highest DISCERN scores, followed by YouTube, TikTok, and Instagram. JAMA compliance was modest overall, with currency being the most frequently satisfied criterion and attribution and disclosure the weakest. Creator type was associated with quality, with healthcare professionals and certified fitness professionals outperforming general users. Engagement metrics did not consistently signal higher quality. On YouTube, higher views and likes were associated with lower DISCERN scores.

Discussion

The findings suggest that platform architecture matters for information quality. Traditional websites retain structural advantages, including greater space for context, attribution, and update history. Short-form social media, by contrast, appears more vulnerable to brevity, limited source disclosure, and engagement-driven ranking. The significant negative within-platform correlations observed on YouTube indicate that popularity should not be treated as a proxy for quality. These results support the need for stronger digital health literacy and platform-level quality safeguards.

Conclusions

Physical activity information quality differs substantially across digital platforms. Traditional websites remain the most reliable source, whereas short-form social media content is more

likely to lack transparency and to provide lower-quality information. Professionals produce stronger content on average, but platform effects remain important. Future interventions should focus on content quality standards, source transparency, and platform-aware health communication strategies.

APPENDIX F: EVIDENCE OF SCIENTIFIC CONFERENCE PRESENTATION

Official conference certificate confirming presentation of the thesis findings, included in accordance with LSU regulation clause 41.2.

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Ademola-John Ayokunle

at 10th April, 2025 presented a scientific report and participated at

international scientific conference „Rehabilitation and Health Promotion: Future Perspectives“ at Lithuanian Sports University (6 hours)

Žinių ir inovacijų perdavimo departamento direktorė *Aida Vengalė*

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APPENDIX G: CONSULTATIONS TIMESHEET

This annex is reserved for the completed consultations timesheet coordinated with the scientific supervisor, as required by LSU regulation clause 12.2.

Insert the signed or official document in this annex before submission.

APPENDIX H: ETHICS PERMISSION OR FORMAL EXEMPTION

This annex is reserved for the official ethics approval, ethics exemption, or Study Programme Committee decision documenting why formal ethics approval was not required, as referenced in LSU regulation clauses 14.2 and 41.3.

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